



Arab Republic of Egypt
Ministry of Education
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MECHANICS

FOR SECOND FORM SECONDARY

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غير مصرح بتداول هذا الكتاب
خارج وزارة التربية والتعليم

مطابع
النهضة



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MECHANICS

*FOR
SECOND FORM SECONDARY*

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غير مصرح بتداول هذا الكتاب خارج وزارة التربية والتعليم

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المقدمة

يسعدنا أن نقدم لأبنائنا طلبة وطلبات الصف الثانى الثانوى هذا الكتاب فى الميكانيكا، وفى ضوء توجيهات السيد الأستاذ / وزير التربية والتعليم.

وقد راعينا فى عرض موضوعاته التأكيد على المفاهيم والمهارات الرياضية التى تساعده الطالب على حل المشكلات معتمداً فى ذلك على المستويات العليا للتفكير وليس الحفظ والاستظهار.

ولم نغفل فى عرض المادة العلمية، ربط ما يدرسه الطالب بواقع حياته اليومية وقد راعينا أيضاً فى عرض الأمثلة والتمارين التدرج من السهل إلى الصعب، ومراعاة الفروق الفردية بين الطلاب.

ويحتوى هذا الكتاب موضوعى الاستاتيكا والديناميكا :

أولاً: الاستاتيكا عرض هذا الجزء فى فصلين، الفصل الأول: «المتجهات والقوى» وهو مقدمة أساسية لدراسة علم الميكانيكا، والفصل الثانى «القوى المستوية المتلاقية فى نقطة» وهو أحد الموضوعات الهامة التى تهتم بالقوى من حيث اتزانها وإيجاد محصلة مجموعة من القوى.

ثانياً: الديناميكا وتعنى دراسة الحركة وفيها يتعرف الطالب على مفاهيم جديدة مثل مفهوم السرعة النسبية والمتوسطة واللحظية كما يتعرف على الحركة الرأسية تحت تأثير عجلة منتظمة أو عجلة الجاذبية الأرضية.

وقد ذيلنا كل فصل بمجموعة من التمارين العامة والكتاب بمجموعة من الاختبارات التى تؤكده على قياس نواتج التعلم المراد تمكين الطالب منها وهى مزودة بإرشادات وإجابات نهائية.

والله من وراء القصد

I. STATICS

CONTENT

<u>Subject</u>	<u>Page</u>
First:- Statics	
Chapter one: Vectors and forces	7
Chapter two: Equilibrium of a body under a coplanar concurrent set of forces.	41
General Exercises	61
 Second: Dynamics	
Chapter one: Relative velocity – Variable motion – Vertical motion under gravity	67 90
Chapter two: Differentiation of vector functions	101
General Exercises	115
Model Tests	120
Answers	127

CHAPTER I

FIRST : VECTORS

Chapter I

FIRST : VECTORS

Scalar Quantity

It is a quantity that completely determined by a single real number which represent the magnitude of the quantity concerned. For example : length, time, mass, volume and temperature are scalar quantities.

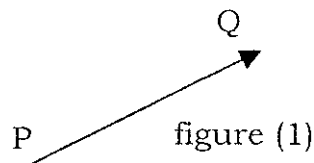
Vector Quantity (The vector)

It is a quantity that completely determined by magnitude and direction. For example : force, translation, velocity and acceleration.

Directed Line Segment

In figure (1), the directed line segment

From P to Q, denoted by $\overrightarrow{PQ}$, is the



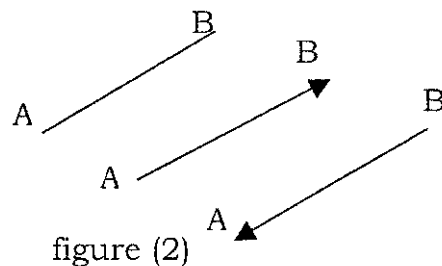
line segment that starts from P and ends at Q.

P is called the initial point and Q is called the terminal point of the directed line segment $\overrightarrow{PQ}$.

In figure (2) :

$\overline{AB} \equiv \overline{BA}$ but

$\overrightarrow{AB} \neq \overrightarrow{BA}$



So, we can say that a directed line segment is determined by three elements, namely :

(1) an initial point (2) a terminal point (3) a direction from the initial point towards the terminal one.

Definition 1

Two directed line segments $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are said to be equivalent to each other if they have equal lengths and have the same direction

In figure (3), the directed line segments $\overrightarrow{AB}$, $\overrightarrow{CD}$ and $\overrightarrow{EF}$ are equivalent

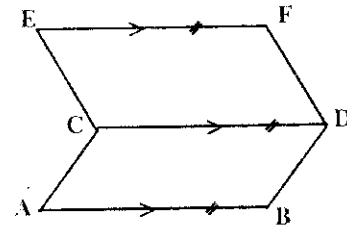


figure (3)

In figure (4), $\overrightarrow{AB}$ and $\overrightarrow{CD}$ have the same direction and have different lengths, so, $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are not

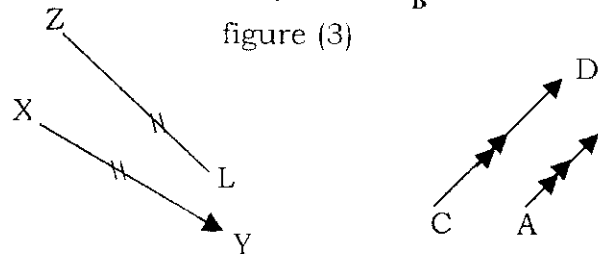


figure (4)

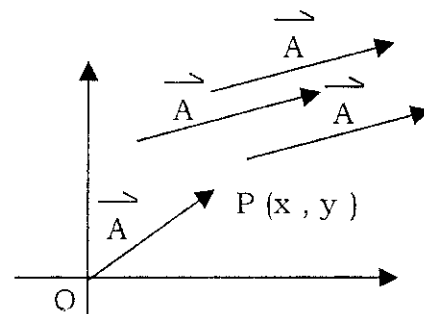
equivalent. $\overrightarrow{XY}$ and $\overrightarrow{ZL}$ have the same lengths, but they have different directions, so, $\overrightarrow{XY}$ and $\overrightarrow{ZL}$ are not equivalent.

The position vector

If $P(x, y)$ is a point in the plane, and O is the origin of a rectangular coordinates system established in the plane, then the directed line segment $\overrightarrow{OP}$ is called the position vector of the point P .

$\overrightarrow{OP}$ is said to be a geometrical representation of the vector $\vec{A} = (x, y)$.

Then any directed line segment equivalent to $\overrightarrow{OP}$ can be considered as a geometrical representation of $\vec{A}$



Definition 2

The two vectors $\vec{A} = (x_1, y_1)$ and $\vec{B} = (x_2, y_2)$
are equal if and only if $x_1 = x_2$ and $y_1 = y_2$

Definition 3

If $\vec{A} = (x_1, y_1)$ and $\vec{B} = (x_2, y_2)$,
then $\vec{A} + \vec{B} = (x_1 + x_2, y_1 + y_2)$

For example, if $\vec{A} = (2, 3)$ and $\vec{B} = (-1, 4)$,

$$\text{then } \vec{A} + \vec{B} = (2-1, 3+4) = (1, 7)$$

Definition 4

If $\vec{A} = (x, y)$ and $K \in \mathbb{R}$
Then $K\vec{A} = K(x, y) = (Kx, Ky)$

In Definition 4 : if $K > 0$, then the two vectors $\vec{A}$ and $K\vec{A}$ are parallel and have the same direction, and if $K < 0$, then $\vec{A}$ and $K\vec{A}$ are parallel, but have opposite directions.

For example in figure (6) :

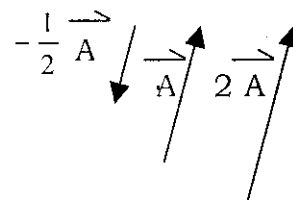
- $\vec{A}$ and $2\vec{A}$ are parallel and

have the same direction

- $\vec{A}$ and $-\frac{1}{2}\vec{A}$ are parallel and have opposite directions. figure (6)

- If $\vec{A} = (4, 6)$, then $2\vec{A} = 2(4, 6) = (8, 12)$, and

$$\frac{1}{2}\vec{A} = \frac{1}{2}(4, 6) = (2, 3)$$



Geometric addition of Vectors

Geometrically, addition of vectors can be illustrated as a sum of displacements. In this sense we have two rules which are the triangle rule and the parallelogram rule.

The Triangle Rule

In figure (7) :

$\vec{DE}$ represents the vector $\vec{A}$, and

$\vec{EF}$ represents the vector $\vec{B}$, then $(\vec{A} + \vec{B})$ is considered as the displacement $\vec{DE}$ followed by the displacement $\vec{EF}$. Since $\vec{DE} + \vec{EF} = \vec{DF}$

$$\therefore \vec{A} + \vec{B} = \vec{DF}$$

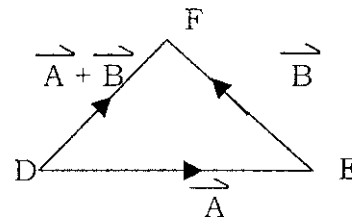


Figure (7)

The Parallelogram Rule

In figure (8) :

If $\vec{A}$ and $\vec{B}$ are represented by two adjacent sides of a

Parallelogram, then we can prove that

$\vec{A} + \vec{B}$ is represented by the diagonal of

this parallelogram $\vec{DF}$.

i.e. $\vec{A} + \vec{B} = \vec{DF}$ [the proof is an exercise for the student].

Remark : $\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$

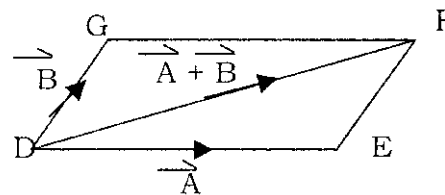


figure (8)

The difference between two vectors (geometrically)

In figure (9) :

If $\vec{DE}$ represents the vector $\vec{A}$, and

$\vec{DF}$ represents the vector $\vec{B}$, then

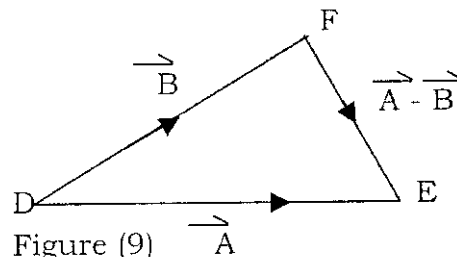


Figure (9)

The difference $\vec{A} - \vec{B}$ is represented
by the directed line segment $\vec{FE}$

[Proof is an exercise for the student]

Notice that the pattern of symbols in the relation :

$$\vec{DE} - \vec{DF} = \vec{FE}$$

Example : ABCD is a quadrilateral, prove that :

$$1) \vec{AB} + \vec{BC} + \vec{CD} = \vec{AD}$$

$$2) \vec{DB} - \vec{CB} = \vec{DA} - \vec{CA}$$

Solution :

$$1) \text{ L.H.S} = (\vec{AB} + \vec{BC}) + \vec{CD}$$

$$= \vec{AC} + \vec{CD}$$

$$= \vec{AD} = \text{R.H.S}$$

$$2) \text{ L.H.S} = \vec{DB} + (-\vec{CB})$$

$$= \vec{DB} + \vec{BC}$$

$$= \vec{DC} \quad (1)$$

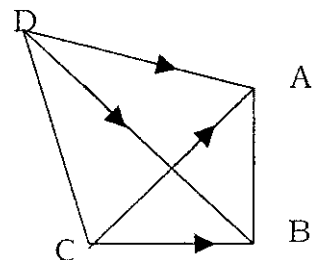
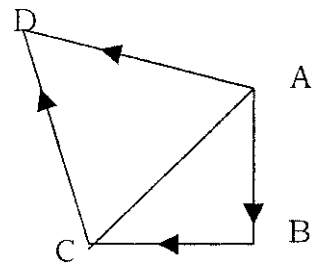
$$\text{R.H.S} = \vec{DA} + (-\vec{CA})$$

$$= \vec{DA} + \vec{AC}$$

$$= \vec{DC} \quad (2)$$

From (1) and (2)

$$\therefore \vec{DB} - \vec{CB} = \vec{DA} - \vec{CA}$$



Vectors and Coordinates

In figure (10) :

The vector $\vec{i} = (1, 0)$ is

represented by the position vector

of the point $(1, 0)$ on the x-axis,

also, the vector $\vec{j} = (0, 1)$ is

represented by the position vector

of the point $(0, 1)$ on the y-axis.

if $\vec{A}$ is represented by $\vec{OP}$, then we write :

$$\vec{A} = \vec{OP} = x\vec{i} + y\vec{j}$$

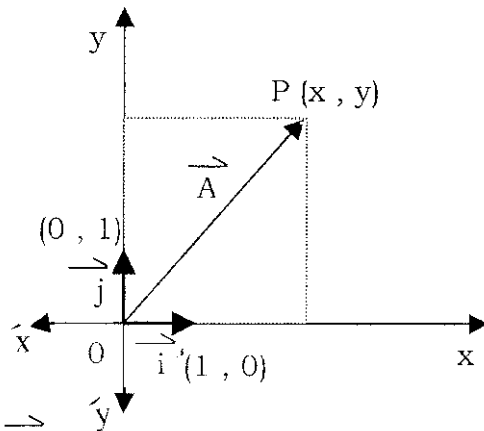


Figure (10)

This result is a direct consequence from the fact that the two vectors $x\vec{i}$ and $y\vec{j}$ are represented by the two position vectors of the two points $(x, 0)$ and $(0, y)$ respectively

So, $\vec{A} = (x, y) = (x, 0) + (0, y)$

$$= x(1, 0) + y(0, 1)$$

$$= x\vec{i} + y\vec{j}$$

The Norm (or the length) of the vector

The norm of the vector $\vec{A}$ is denoted by $\|\vec{A}\|$ where

$$\|\vec{A}\| = \sqrt{x^2 + y^2}$$

For example : If $\vec{A} = (3, 4)$, then $\|\vec{A}\| = \sqrt{9 + 16} = 5$,

And if $\vec{B} = (-1, 3)$, then $\|\vec{B}\| = \sqrt{1 + 9} = \sqrt{10}$

The unit vector

A vector whose norm is 1 is called the unit vector

In figure (10) : $\vec{i} = (1, 0)$ and $\vec{j} = (0, 1)$ are two unit vectors, and they are called basic unit vectors.

The Zero Vector

It is denoted by $\vec{0} = (0, 0)$, and we notice that $\vec{A} + \vec{0} = \vec{A}$

Exercise 1

(1) In any triangle ABC, prove that $\vec{AB} + \vec{BC} + \vec{CA} = \vec{0}$

(2) ABCD is a quadrilateral in which : $2\vec{BC} = 5\vec{AD}$.

Prove that : $2\vec{AB} - 2\vec{DC} = 3\vec{DA}$

(3) In the parallelogram ABCD, prove that :

(i) $2\vec{AM} = \vec{AB} + \vec{AD}$, where M is the point of intersection of its diagonals.

(ii) $\vec{PD} + \vec{PB} = \vec{PA} + \vec{PC}$, where P is any point in the plane.

(4) ABCD is a parallelogram in which A (1 , 2), B (5 , 7) and C (8 , -3).

Find D.

(5) Find the norm of each of the following vectors :

$\vec{A} = (2 , -3)$, $\vec{B} = (3 , 2)$, $\vec{C} = (0 , -3)$ and $\vec{D} = (4 , 0)$

(6) If $\vec{A} = (3 , 1)$ and $\vec{B} = (1 , 2)$, find $\|\vec{A} - 2\vec{B}\|$

SECOND : THE FORCES

The Force :

The force is defined as the effect of a natural body upon another one. The natural body we mean that one consisting of matter with non vanishing volume.

From this point of view bodies can be classified into :

Rigid bodies and deformable bodies :

A rigid body is one in which every pair of points is considered to be at a constant distance apart at all times, regardless of the type acting force upon it. Accordingly, a rigid body is never deformed but a deformable body is easily deformed by any acted force. In fact, there is no completely rigid real solid, but the concept of a rigid body is very useful in representing a some mechanical problems and consequently in solving it.

As an example of a rigid body we mention the solid bodies and of deformable bodies we mention strings, clays, liquids and gases of all kinds.

The fundamental concepts underlying mechanics is that of (Force).

Mechanics

In dynamics, force is defined as the influence that changes or tries to change the state of a body being at rest or uniform motion.

In statics, the interest of motion study does not arise except due to the unfulfilment of the conditions of equilibrium.

The effect of the force that a body acts upon another changes form state into another, this effect could be tension, pressure, or the attraction force of bodies to each other (bodies are not in contact).. etc.

As example of the last kind of forces we mention the earth attraction force (gravitational force) to bodies which is called (the weight of the body) the attraction force between the sun and the planets, the attraction and repulsion forces among bodies or magnits, and nuclear forces are all examples of different types of forces.

There exists another kind of forces arises due to the touch between any two bodies it is called (the reaction), for example in case of placing a body on a table, the body acts on a table some force. This force is not generated by the weight of the body but its magnitude depends on it, and, in turn, the table acts on the body by another force.

Properties of Force :

The effect of any force depends on three factors :

1. The magnitude of the force.
2. The direction of force.
3. The point of action, hence its line of action.

POSTULATE

The force is a vector which passes through a given point or acts in a given straight line.

This postulate is one of the fundamental postulates in mechanics whose validity is proved experimentally.

The Vector of the Force :

It is known that a vector is well defined by :

1. Magnitude.
2. Direction.

Therefore, the force is one of the familiar examples of the vectors. Thus, we can say that the force is a vector which passes through a given point or acts in a given straight line.

The magnitude of the force is measured (in the french system) by the unit of (gramme weight). Recently international unit is used (The newton).

The newton is defined as the force which acts upon a mass of 1 kg, it generates acceleration 1m/sec.^2 .

There are many other units to measure the magnitude of the force such as the kilogram weight (kg.wt) and (the dyne).

$$1 \text{ kg.wt} \approx 9.8 \text{ newtons approximately.}$$

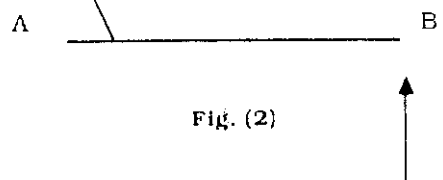
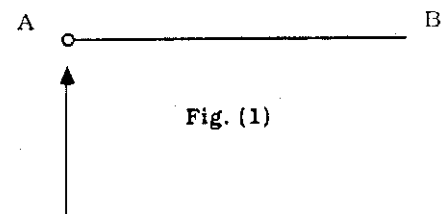
$$= 1000 \text{ gm.wt.}$$

$$1 \text{ dyne} = \frac{1}{100000} \text{ newton} = 10^{-5} \text{ newton.}$$

The direction of the force is defined by the direction of the vector which represents this force. If we restrict our attention only to the forces which are represented by coplaner vectors the direction of the force could be determined by the polar angle of the vector of the force.

The concepts of the point of action of the force could be clarified by using the following example which can be easily imagined by the student.

Let a rod AB placed on a horizontal smooth plane and hinged at its end A such that it can be rotated horizontally



1. Act by a perpendicular force to the rod at A, we notice that the rod will remain at rest. (Fig. 1).
2. Act by the same force at B, the rod will rotate about A. (fig(2))

This simple experiment shows that the action of the force depends on the point of its action. The straight line passing thorough the point of action parallel to the direction of the force is called (the line of action of the force) e.g. the line of action of the tension of a rope is the rope itself, the line of action of the weight of a body is the vertical line passing through its centre gravity.

From now on let us denote force of magnitude (F), by $\vec{F}$ or in the form of a quantity between brackets contain magnitude and direction (e.g.) the force shown in (Fig. 3) is :

$$\vec{F} = (100 \text{ newtons}, 33^\circ).$$

Example 1.

In Fig. (3) a force = 100 newtons

Scale 1 cm/20 newtons

direction 33° with the fixed ray $\vec{OX}$ anticlockwise direction.

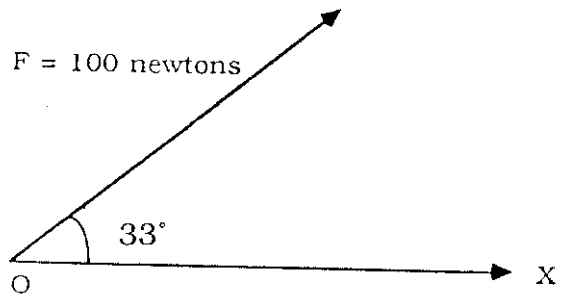


Fig. (3)

COMPOSITION OF FORCES

The resultant of two forces :

We can obtain a single force has the same effect as a system of forces (two or more forces).

This single force is called the resultant force to the system of forces.

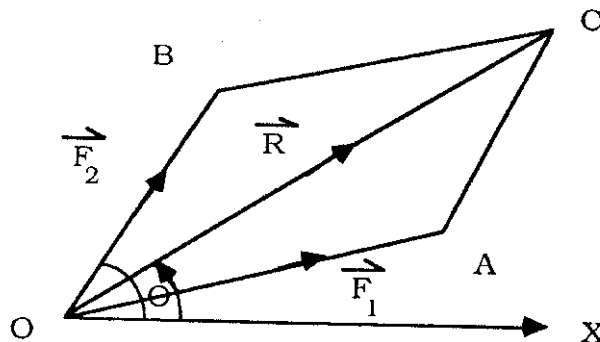


Fig. (10)

To find the resultant of two forces $\vec{F}_1$, $\vec{F}_2$ meeting at a point. Draw the vector $\vec{OA}$ representing $\vec{F}_1$ and the vector $\vec{OB}$ to

represent $\vec{F}_2$, then draw $\vec{AC}$ parallel to $\vec{OB}$ and $\vec{BC}$ parallel to $\vec{OA}$ to meet at C. (Fig. 10).

$\vec{OC}$ is one of the diagonals of the parallelogram OACB. We know from our study of addition of two vectors that the vector $\vec{OC}$ is the sum of the two vectors $\vec{OA}$, $\vec{OB}$ and since this vector starts by the intersection point O of the lines of action of the two forces represented by the two straight directed segments $\vec{OA}$, $\vec{OB}$, then the vector $\vec{OC}$ represents the sum of $\vec{F}_1$, $\vec{F}_2$.

The magnitude of this force can be found by comparing the length of $\vec{OC}$ with the scale used in determining the lengths of $\vec{OA}$, $\vec{OB}$. The direction of this force is given by the measure of angle between $\vec{OC}$ and the fixed direction $\vec{OX}$ in the plane of $\vec{OA}$, $\vec{OB}$.

The single force denoted by $\vec{R}$ makes the same effect made by the two forces $\vec{F}_1$, $\vec{F}_2$, so $\vec{R}$ is called the resultant of the two forces $\vec{F}_1$, $\vec{F}_2$.

As in case of addition of two vectors, this rule is called "The rule of parallelogram of forces."

The statement of the rule :

"If two forces meeting at a point be represented in magnitude and direction by two sides of a parallelogram meeting at a point, then their resultant is represented in magnitude and direction by the diagonal of the parallelogram which meets the two sides at the same

Remark :

In this rule, the two forces meet at a point and since the force must act upon a body, then the point at which the two forces meet must be a point in the body, there is no objection for the body to be a particle.

Let two forces acting upon a body represented in magnitude, direction and line of action by the two vectors $\vec{AB}$, $\vec{BC}$. (Fig. 11).

The resultant of these two vectors is represented by $\vec{AC}$ but the line of action of the resultant must pass through B which is the only point acted upon by the two forces.

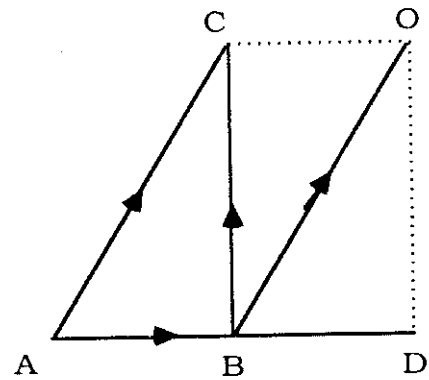


Fig. (11)

$\therefore \vec{BO} = \vec{AC}$, then the resultant is represented by the vector $\vec{BO}$.

This agrees with the rule of the parallelogram, because by completing the parallelogram BDOC, we find that the point D lies on $\vec{AB}$, $D \in \vec{AB}$ and $AB = BD$.

$\vec{AB} = \vec{BD}$ and this means that :

$$\vec{BO} = \vec{BD} + \vec{BC} = \vec{AB} + \vec{BC}.$$

The parallelogram of forces rule is a fundamental postulate in mechanics.

This means that the laws of addition of vectors are also correct for forces.

To find the result of two forces analytically :

Let the two forces $\vec{F}_1$, $\vec{F}_2$ meeting at O and α is the measure of the angle between the direction of the two forces. (Fig. 12).

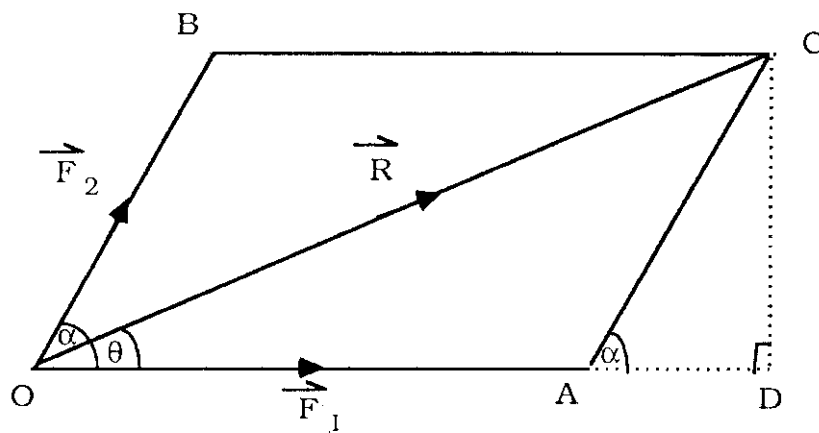


Fig. (12)

$\vec{OA}$ represents $\vec{F}_1$, $\vec{OB}$ represents $\vec{F}_2$

$\vec{OC}$ represents $\vec{R}$, (the resultant of two forces). $\vec{OC}$ is the diagonal of parallelogram constructed of the two adjacent sides $\vec{OA}$, $\vec{OB}$.

θ is the measure of the angle of inclination of the resultant $\vec{R}$ with $\vec{F}_1$.

From the cosine law we can get the resultant of two forces $\vec{F}_1$, $\vec{F}_2$ in magnitude and direction by the two following relations.

$$R = \sqrt{F_1^2 + F_2^2 + 2F_1 F_2 \cos \alpha}$$

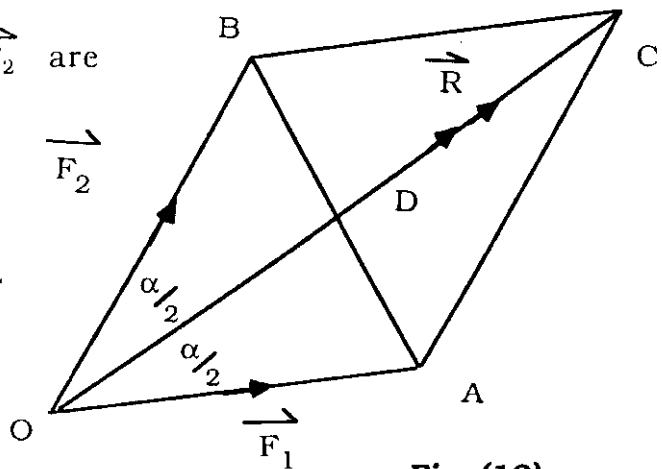
$$\tan \theta = \frac{F_2 \sin \alpha}{F_1 + F_2 \cos \alpha}$$

Special cases :

1. If the two forces $\vec{F}_1, \vec{F}_2$ are equal in magnitude :

$$F_1 = F_2 = F$$

$$\begin{aligned} \therefore R^2 &= F^2 + F^2 + 2F^2 \cos \alpha \\ &= 2F^2 (1 + \cos \alpha) \\ &= 4F^2 \cos^2 \frac{\alpha}{2} \end{aligned}$$

**Fig. (13)**

$$\therefore R = 2F \cos \frac{\alpha}{2} \qquad \theta = \frac{\alpha}{2}$$

The resultant bisects the angle between the two forces $\vec{F}_1, \vec{F}_2$ (Fig. 13).

Another Proof :

$$\begin{aligned} \cos \frac{\alpha}{2} &= \frac{OD}{OA} = \frac{\frac{1}{2} R}{F} \\ \frac{1}{2} R &= F \cos \frac{\alpha}{2} \\ R &= 2 F \cos \frac{\alpha}{2} \end{aligned}$$

2. If the measure of the angle

between $\vec{F}_1, \vec{F}_2$ is 90° (Fig. 14).

$$\alpha = 90^\circ$$

$$\therefore \cos 90^\circ = 0$$

$$\therefore R^2 = F_1^2 + F_2^2$$

$$\tan \theta = \frac{F_2}{F_1}$$

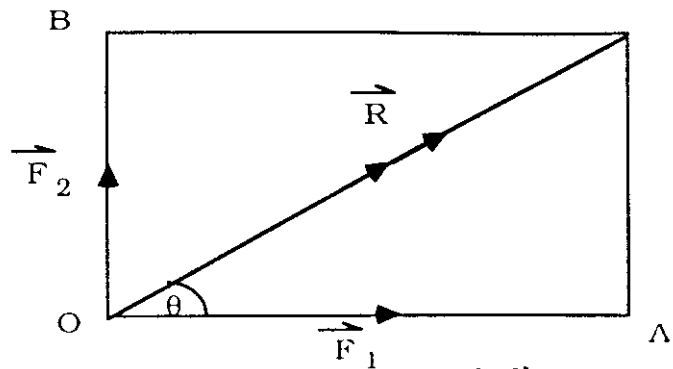


Fig. (14)

3. If the two forces $\vec{F}_1, \vec{F}_2$ have the same line of action.

a) $\vec{F}_1, \vec{F}_2$ have the same direction.

$\therefore R = F_1 + F_2$ and $\vec{R}$ is in the same direction of the two forces.

b) $\vec{F}_1, \vec{F}_2$ are in two opposite directions.

$\therefore R = |F_1 - F_2|$ and $\vec{R}$ is in the direction of the greatest force.

4. If $\vec{F}_1, \vec{F}_2$ are equal in magnitude, opposite in direction and then lines of action are on the same straight line (Fig. 15).

$$F_1 = F_2 = F.$$

$$\alpha = 180^\circ$$

$$\therefore \cos \alpha = -1$$

$$R^2 = (F)^2 + (F)^2 - 2(F)^2$$

$$= 0$$

$$\therefore R = 0$$

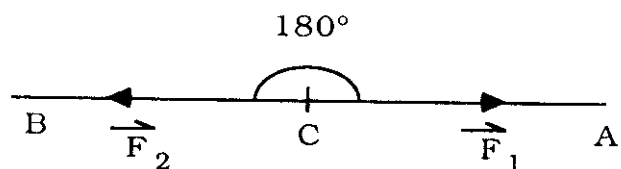


Fig. (15)

$\therefore$ The resultant is the Zero vector.

Example (1) :

The forces $5, 5\sqrt{2}$ newtons act at a point, include an angle of measure 45° . Find their resultant in magnitude and direction.

Solution :

$$F_1 = 5 \text{ newtons}$$

$$F_2 = 5\sqrt{2} \text{ newtons}$$

$$\alpha = 45^\circ$$

$$\cos \alpha = \frac{1}{\sqrt{2}}$$

$$\begin{aligned} R^2 &= (5)^2 + (5\sqrt{2})^2 + 2 \times \\ &\quad 5 \times 5\sqrt{2} \times \frac{1}{\sqrt{2}} \\ &= 25 + 50 + 50 \\ &= 125 \end{aligned}$$

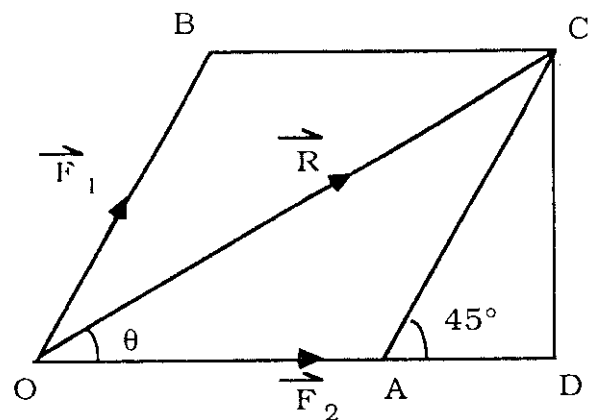
$$\therefore R = 5\sqrt{5} \text{ newtons}$$

$$\tan \theta = \frac{F_2 \sin \alpha}{F_1 + F_2 \cos \alpha}$$

$$= \frac{5\sqrt{2} \times \frac{1}{\sqrt{2}}}{5 + 5\sqrt{2} \times \frac{1}{\sqrt{2}}}$$

$$= \frac{1}{2}$$

$$\theta = 26^\circ 34' \text{ (Fig. 16).}$$

**Fig. (16)**

Example (2) :

Find the magnitude and the direction of the resultant of the two forces $\vec{F}_1$, $\vec{F}_2$. If $F_1 = 750$ newtons, $F_2 = 500$ newtons and the measure of the angle between them is 60° .

Solution :

$$F_1 = 750 \text{ newtons.}$$

$$F_2 = 500 \text{ newtons.}$$

$$\alpha = 60^\circ.$$

$$\begin{aligned} R^2 &= (750)^2 + (500)^2 + 2 (750) (500) \cos 60^\circ. \\ &= 562500 + 250000 + 375000 \\ &= 1187500 \end{aligned}$$

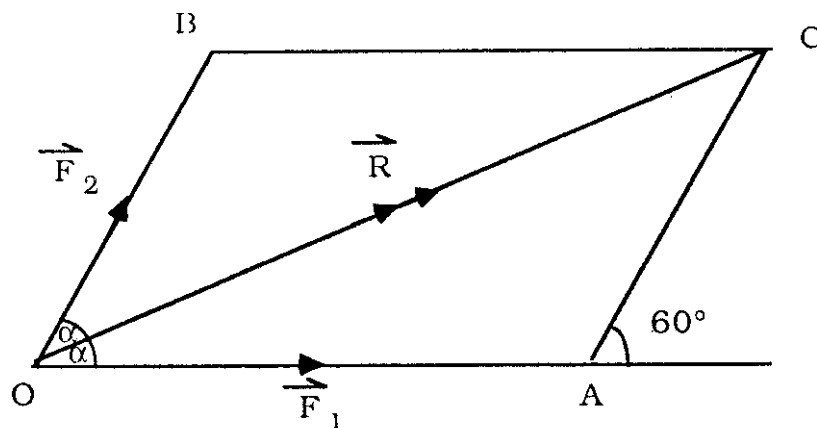


Fig. (17)

$$\therefore R = 1089.72 \text{ newtons}$$

$$\begin{aligned} \tan \theta &= \frac{F_2 \sin \alpha}{F_1 + F_2 \cos \alpha} \\ &= \frac{500 \sin 60^\circ}{750 + 500 \cos 60^\circ} \\ &= \frac{250 \sqrt{3}}{750 + 250} = \frac{\sqrt{3}}{4} = 0.433 \end{aligned}$$

$$\therefore \theta = 23^\circ 25' \text{ (Fig. 17).}$$

Example (3) :

If the measure of the angle between two forces is 90° , their resultant is $\sqrt{10}$ newtons, and if the measure is 60° , their resultant is $\sqrt{13}$ newtons. Find the magnitude of each force.

Solution :

Let F_1 , F_2 be the magnitudes of the two forces, α is the measure of the angle between them.

$$R^2 = F_1^2 + F_2^2 + 2 F_1 F_2 \cos \alpha$$

In the 1st case : $\alpha = 90^\circ$, $R = 10$.

$$10^2 = F_1^2 + F_2^2 \dots\dots\dots (1)$$

In the 2nd case : $\alpha = 60^\circ$, $R = 13$

$$\begin{aligned} 13^2 &= F_1^2 + F_2^2 + 2 F_1 F_2 \cos 60^\circ \\ &= F_1^2 + F_2^2 + F_1 F_2 \dots\dots\dots (2) \end{aligned}$$

From (1) and (2) :

$$F_1 \cdot F_2 = 3$$

$$\begin{aligned} \therefore (F_1 + F_2)^2 &= F_1^2 + 2 F_1 F_2 + F_2^2 \\ &= (F_1^2 + F_2^2) + 2 F_1 F_2 = 10 + 6 = 16 \end{aligned}$$

$$\therefore F_1 + F_2 = 4 \dots\dots\dots (3)$$

Also :

$$(F_1 - F_2)^2 = F_1^2 - 2 F_1 F_2 + F_2^2$$

$$\begin{aligned}
 &= (F_1^2 + F_2^2) - 2 F_1 F_2 \\
 &= 10 - 6 \\
 &= 4
 \end{aligned}$$

$$F_1 - F_2 = 2 \quad (\text{Supposing that } F_1 > F_2) \dots\dots\dots (4)$$

From (3) and (4) :

$$\text{by adding} \quad 2 F_1 = 6 \quad \therefore F_1 = 3 \text{ newtons}$$

$$\text{by subtraction} \quad 2 F_2 = 2 \quad \therefore F_2 = 1 \text{ newton}$$

Example (4) :

Two forces of magnitudes F , $2F$ act at the point O . Find the measure of the angle between them if their resultant is perpendicular to the first force.

Solution :

$$\begin{aligned}
 \tan \theta &= \frac{F_2 \sin \alpha}{F_1 + F_2 \cos \alpha} \\
 \therefore \theta &= 90^\circ. \\
 \therefore \cot \theta &= 0 \\
 (\text{i.e.}) F_1 + F_2 \cos \alpha &= 0 \\
 \therefore F + 2 \cos \alpha &= 0 \\
 \therefore \cos \alpha &= -1/2 \\
 \therefore \alpha &= 120^\circ
 \end{aligned}$$

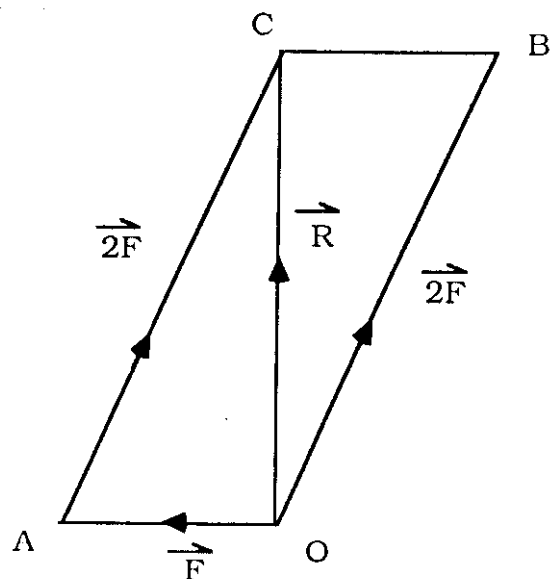


Fig. (18)

Another solution :

From the figure, if $\overrightarrow{OA}$ represents the force whose magnitude is F , $\overrightarrow{OC}$ represents $\vec{R}$, then $\overrightarrow{AC}$ represents the force $2F$ in magnitude and direction.

$\therefore$ AC equals twice AO

(i.e.) The measure of the angle $\angle ACO = 30^\circ$.

$\therefore$ The measure of the angle between the two forces =
 $90^\circ + 30^\circ = 120^\circ$.

Resolution of a Force into two Components

We have already seen how to find the resultant of two given forces $\vec{F}_1$, $\vec{F}_2$ met at one point by using the rule of the parallelogram of forces.

Now we are going to see how to find the two forces if their resultant $\vec{R}$ is given, this reverse operation is called resolution of the force $\vec{R}$ into two components. From our previous study of vectors we know that we can perform this operation by infinite number of ways, that is because we can draw an infinite number of parallelograms have one diagonal. But if the direction of the two forces $\vec{F}_1$, $\vec{F}_2$ are given the result of resolution is unique.

Suppose that the force $\vec{R}$ which is required to be resolved into two components is represented by the vector $\vec{OC}$.

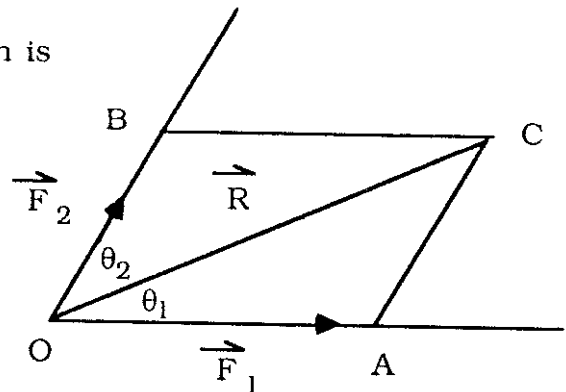


Fig. (19)

Let θ_1 and θ_2 be the measures of the angles of inclination of F_1 and F_2 to $\vec{OC}$ respectively (Fig. 19).

By using the sine rule :

$$\frac{OA}{\sin \theta_2} = \frac{AC}{\sin \theta_1} = \frac{OC}{\sin (180^\circ - \theta_1 - \theta_2)} = \frac{OC}{\sin (\theta_1 + \theta_2)}$$

$$\therefore \frac{F_1}{\sin \theta_2} = \frac{R}{\sin (\theta_1 + \theta_2)}, \quad \frac{F_2}{\sin \theta_1} = \frac{R}{\sin (\theta_1 + \theta_2)}$$

$\therefore$ The magnitude of R and the measure of θ_1 and θ_2 are given.

$$\therefore F_1 = \frac{R \sin \theta_2}{\sin (\theta_1 + \theta_2)} \qquad F_2 = \frac{R \sin \theta_1}{\sin (\theta_1 + \theta_2)}$$

Example (1) :

Resolve a force 100 newtons in two directions the first inclines by 30° to the force and the other by 45° in the other side of the given force.

Solution :

Let us consider a scale 1 cm : 20 newtons Draw $\overrightarrow{OC}$ to represent the given force from the point O draw two rays from $30^\circ, 45^\circ$ with $\overrightarrow{OC}$, then draw from C two other rays parallel to the previous two forces to intersect them at A, B (Fig. 20), we get a parallelogram $OACB$, then $\overrightarrow{OA}$ represents the first component $\overrightarrow{F_1}$ and $\overrightarrow{OB}$ represents the second component $\overrightarrow{F_2}$.

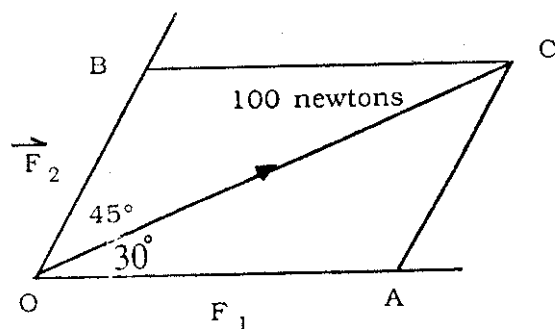


Fig. (20)

By direct measure of the drawing, we find that :

$$OA = 3.7 \text{ cm}$$

$$OB = 2.6 \text{ cm.}$$

$$F_1 = 74 \text{ newtons}$$

$$F_2 = 52 \text{ newtons.}$$

Another Solution :

Applying the sine rule

$$\frac{F_1}{\sin 45^\circ} = \frac{F_2}{\sin 30^\circ} = \frac{100}{\sin 105^\circ}$$

$$\therefore F_1 = \frac{100 \sin 45^\circ}{\sin 105^\circ} = \frac{100 \times 0.707}{0.96593} \approx 73.2 \text{ newtons.}$$

$$F_2 = \frac{100 \sin 30^\circ}{\sin 105^\circ} = \frac{50}{0.96593} \approx 51.76 \text{ newtons.}$$

Resolution of Force into Two Perpendicular Directions

If $\vec{OA}$, $\vec{OB}$ are perpendicular, then by applying the sine rule to the right-angled triangle OAC (Fig. 21) we get :

$$R = \frac{F_1}{\sin(90^\circ - \theta)} = \frac{F_2}{\sin \theta} = \frac{R}{\sin 90^\circ}$$

(i.e.) :

$$\frac{F_1}{\cos \theta} = \frac{F_2}{\sin \theta} = \frac{R}{1}$$

$$\therefore F_1 = R \cos \theta$$

$$F_2 = R \sin \theta$$

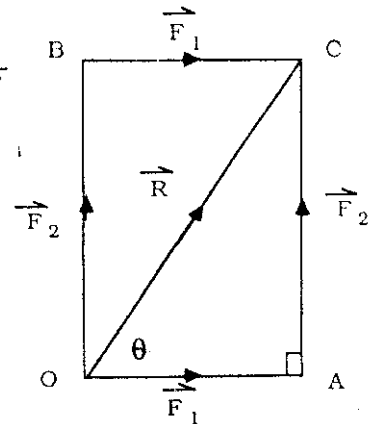


Fig. (21)

The component $\vec{F}_1$ is called sometimes “the projection of $\vec{R}$ to the direction $\vec{OA}$ and $\vec{F}_2$ projection of $\vec{R}$ to the direction of $\vec{OB}$.”

Example (2):

Resolve the force 100 newtons in two perpendicular directions one of them is 30° with the force.

Solution :

In this case the parallelogram is a rectangle, its diagonal represents the force 100 newtons, its adjacent sides form 30° and 60° with the line of action of force.

$$F_1 = 100 \cos 30^\circ = 86.6 \text{ newtons.}$$

$$F_2 = 100 \sin 30^\circ = 50 \text{ newtons.}$$

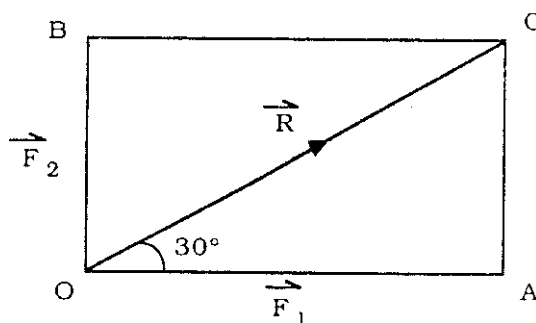


Fig. (22)

Exercise 2

1. Two forces act at a point, α is the angle between them is and $\tan \alpha = -\frac{1}{\sqrt{3}}$. If their resultant is perpendicular to the smaller force, and the greater force is 30 newtons, find the magnitude of the other force and the resultant.
2. Resolve a horizontal force 40 newtons into two perpendicular directions one of them inclines by 30° with the horizontal and upwards.
3. Resolve a force 500 newtons in two directions the first inclines by 45° to the force and the other by 60° in the other side of the given force.
4. Two forces 2, F newtons, the angle between them is of measure 120° . Find F in each of the following cases :
 - (i) The resultant = F.
 - (ii) The resultant is perpendicular to the 2nd force.
 - (iii) The resultant incline by 45° to the 2nd force.
 - (iv) The resultant bisects the angle between the forces.
5. Two forces $\vec{F}_1, \vec{F}_2$ meet at a point, their resultant is R, the angle between them is of measure 120° . If the direction of $\vec{F}_2$ is reversed the resultant will be $R\sqrt{3}$. Prove that $F_1 = F_2$ and the resultant in the first case is perpendicular to the second case.

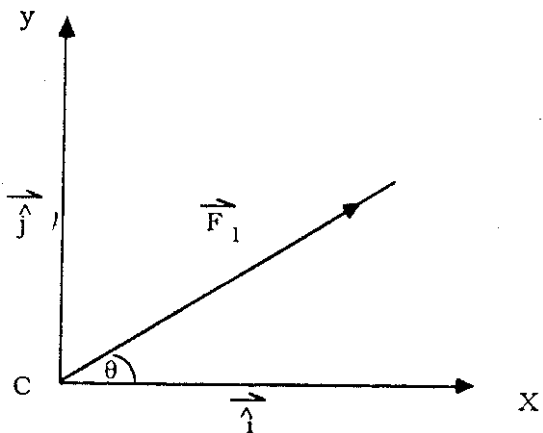
6. A body of weight 20 newtons is placed on an inclined plane of inclination 30° to the horizontal. Find the component of the weight in the direction of the plane and the normal to it.
7. A force of magnitude 100 gm. wt acts in the diections North West, find its components due North and West.

Determining the resultant of a set of concurrent forces.

Suppose that the system of the coplanar forces $\vec{F}_1, \vec{F}_2, \dots$ Met at a point O.

We have seen that any force $\vec{F}$ could be resolved into two perpendicular components.

Let these two directions be $\vec{OX}, \vec{OY}$, the cartesian axes, and $\theta_1, \theta_2, \theta_3, \dots$ are the polar angles of the forces $\vec{F}_1, \vec{F}_2, \vec{F}_3, \dots$ respectively.



Then :

$$\vec{F}_1 = F_1 \cos \theta_1 \hat{i} + F_1 \sin \theta_1 \hat{j}$$

Where $\hat{i}$ and $\hat{j}$ are the fundamental unit vectors respectively.

$F_1 \cos \theta_1$ is called the algebraic component in the direction of $\overrightarrow{OX}$,

$F_2 \sin \theta_1$ is called the algebraic component in the direction of $\overrightarrow{OY}$.

Similarly :

$$\overrightarrow{F_2} = \overrightarrow{F_2} \cos \theta_2 \hat{i} + \overrightarrow{F_2} \sin \theta_2 \hat{j}$$

and so on

$$\therefore \overrightarrow{R} = \overrightarrow{F_1} + \overrightarrow{F_2} + \dots + \overrightarrow{F_n}$$

$$\begin{aligned} \therefore \overrightarrow{R} &= (F_1 \cos \theta_1 \hat{i} + F_1 \sin \theta_1 \hat{j}) + \\ &\quad (F_2 \cos \theta_2 \hat{i} + F_2 \sin \theta_2 \hat{j}) + \dots \\ &= (F_1 \cos \theta_1 + F_2 \cos \theta_2 + \dots + F_n \cos \theta_n) \hat{i} + \\ &\quad (F_1 \sin \theta_1 + F_2 \sin \theta_2 + \dots + F_n \sin \theta_n) \hat{j} \\ \text{or } \overrightarrow{R} &= \left(\sum_{r=1}^n F_r \cos \theta_r \right) \hat{i} + \left(\sum_{r=1}^n F_r \sin \theta_r \right) \hat{j} \end{aligned}$$

Where :

$\sum_{r=1}^n F_r \cos \theta_r$ is the algebraic sum of the components in the $\overrightarrow{OX}$ direction. $\sum_{r=1}^n F_r \sin \theta_r$ is algebraic sum of the components in the $\overrightarrow{OY}$ direction. If we denote these two sums by X and Y respectively, then :

$$X = \sum_{r=1}^n F_r \cos \theta_r, \quad Y = \sum_{r=1}^n F_r \sin \theta_r$$

hence we can express the vector of the resultant $\overrightarrow{R}$ as follows :

$$\overrightarrow{R} = X \hat{i} + Y \hat{j}.$$

Remark :

$\therefore$ The resultant has two components X, Y in the directions $\vec{OX}, \vec{OY}$ respectively, if R is the magnitude of the vector of the resultant and α is the measure of the angle between R and $\hat{i}$ then we can deduce :

$$R = \sqrt{X^2 + Y^2}$$

$$\tan \alpha = \frac{Y}{X}$$

The last relation determine two values for the angle α : $\alpha, 180 + \alpha$ (Say) and we choose one of these two values according to the signs of X and Y (Fig. 24).

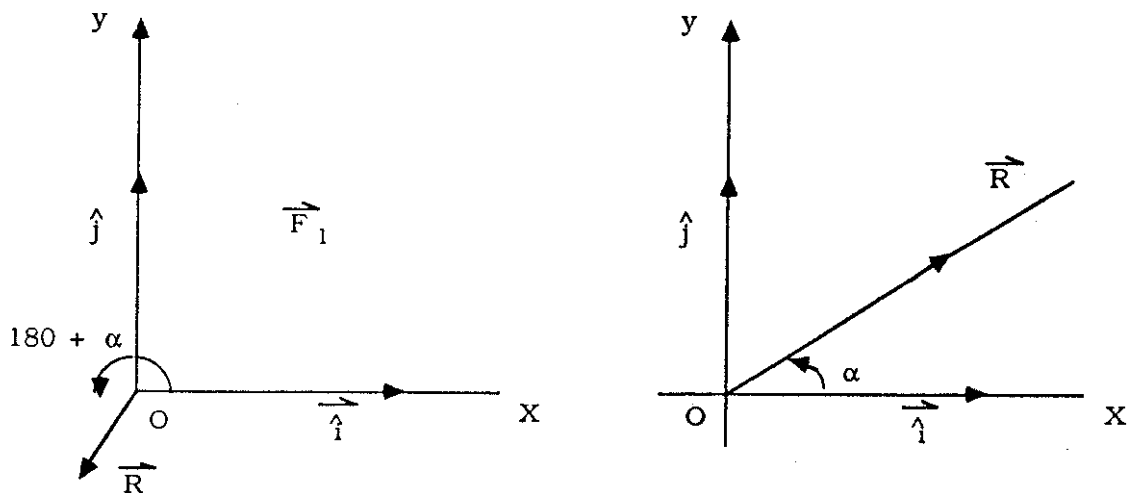


Fig. (24)

Example (1) :

Referring Fig. (25) find the resultant of the system by using the rules of the perpendicular resolution.

Solution :

$$\begin{aligned}\vec{F}_1 &= (50 \cos 30^\circ, 50 \sin 30^\circ) \text{ newtons} \\ &= (25 \sqrt{3}, 25)\end{aligned}$$

$$\vec{F}_2 = (0, 60) \text{ newtons}$$

$$\vec{F}_3 = (-80, 0)$$

$$\begin{aligned}\therefore \vec{R} &= (25 \sqrt{3} - 80, 25 + 60) \\ &= (-36.7, 85).\end{aligned}$$

$$\begin{aligned}\therefore R &= \sqrt{(36.7)^2 + (85)^2} \\ &= 92.58 \text{ newtons}\end{aligned}$$

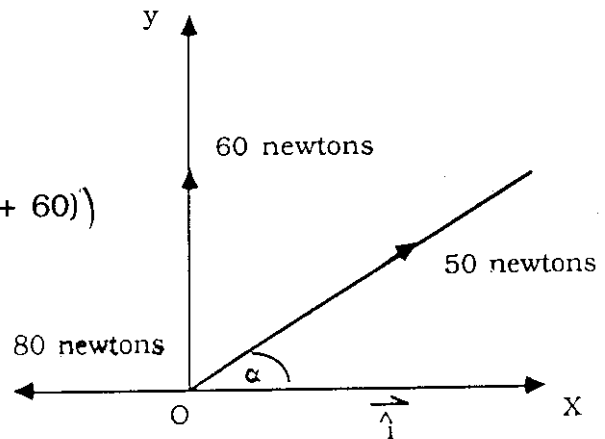


Fig. (25)

$$\begin{aligned}\therefore \tan \alpha &= \frac{Y}{x} = \frac{85}{36.7} = 2.3160 \\ \therefore \alpha &= 113^\circ 20'\end{aligned}$$

Example (2) :

Three forces $2F$, $3F$ and $4F$ newtons act at a particle in directions parallel to the sides of an equilateral triangle taken in the same cyclic order. Find the magnitude and the direction of the resultant.

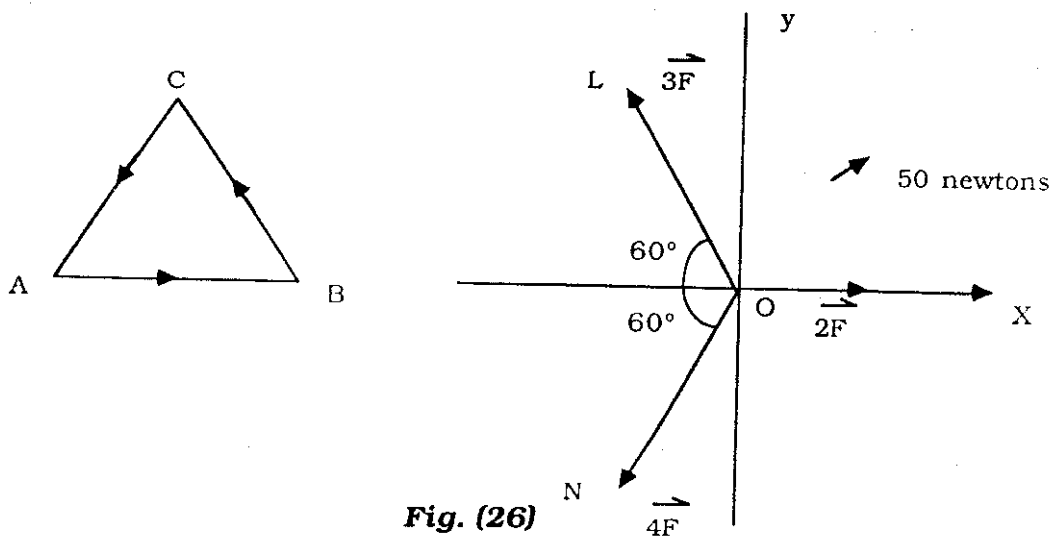


Fig. (26)

Let the forces act in the directions $\vec{OX}$, $\vec{OL}$ and $\vec{ON}$ which are parallel to the sides $\vec{AB}$, $\vec{BC}$ and $\vec{CA}$ respectively in the equilateral triangle ABC (Fig. 26).

Resolve in the perpendicular directions $\vec{OX}$ and $\vec{OY}$.

The algebraic sum of the components in the direction of $\vec{OX}$

$$X = 2F + 3F \cos 120^\circ + 4F \cos 240^\circ.$$

$$= 2F - \frac{3F}{2} - \frac{4F}{2}$$

$$= -\frac{3F}{2} \text{ newtons}$$

The algebraic sum of the components in the $\vec{OY}$ direction is :

$$Y = 3F \sin 120^\circ + 4F \sin 240^\circ$$

$$= 3F \times \frac{\sqrt{3}}{2} + 4F \left(-\frac{\sqrt{3}}{2}\right)$$

$$\therefore Y = -F \cdot \frac{\sqrt{3}}{2} \text{ newtons}$$

$$R = \sqrt{X^2 + Y^2} = \sqrt{\left(-\frac{3}{2}F\right)^2 + \left(-\frac{F\sqrt{3}}{2}\right)^2}$$

$$= F\sqrt{3} \text{ newtons}$$

and the direction of the resultant makes an angle where:

$$\tan \alpha = \frac{Y}{X} = \frac{-\sqrt{3}/2F}{-3/2F} = \frac{1}{\sqrt{3}}$$

$$= 30^\circ \text{ or } 210^\circ$$

$\therefore$ X and Y are negatives

$$\therefore \alpha = 210^\circ$$

Exercise (3)

1. Three coplanar forces of magnitudes 1, 2, $\sqrt{3}$ newtons act at M, their directions are $\overrightarrow{MA}$, $\overrightarrow{MB}$ and $\overrightarrow{MC}$ respectively where $m(\angle AMB) = 60^\circ$, $m(\angle AMC) = 90^\circ$, Find the resultant.
2. Three equal Coplanar forces act on a particle, the line of action of the second force form with the first and the third two angles of measure 60° . Find their resultant.
3. The forces 5, 10, $15\sqrt{3}$, 20 newtons act at a particle, the measure of the angle between the first two forces is 60° , between the second and the third force is 90° , between the third and the fourth is 150° . Find their resultant.
4. Find the resultant of the coplanar forces meet at a point : 5 newtons towards North, 6 newtons towards East, 4 newtons 30° South of West.
5. ABCDEO is regular hexagon. Forces of magnitudes 2, 3, 4, 5, 6 newtons act A along $\overrightarrow{AB}$, $\overrightarrow{AC}$, $\overrightarrow{AD}$, $\overrightarrow{AE}$, $\overrightarrow{AO}$ respectively. Find their resultant.
6. Forces of magnitudes 2, $3\sqrt{2}$, $2\sqrt{3}$ and $\sqrt{3}$ newtons act at a point. If the measure between the first force and the second force is 45° , the measure between the second and the third is 105° and the measure between the third and the fourth is 120° , find the resultant of these forces.

CHAPTER II

EQUILIBRIUM OF COPLANAR FORCES MEETING AT A POINT

We have already discussed the equilibrium of a particle under the action of two forces and we concluded that these two forces must be equal in magnitude, opposite in direction and have the same line of action.

Now, we are going to discuss the equilibrium of three coplanar forces all their lines of action meet at one point.

EQUILIBRIUM OF A BODY UNDER THE ACTION OF TWO FORCES

Rule:

If a body is in equilibrium under the action of two forces only then :

1. The two forces are equal in magnitude.
2. They are opposite in direction.
3. Their lines of action are on the same straight line.

Experiment :

We can verify this rule by the following experiment :

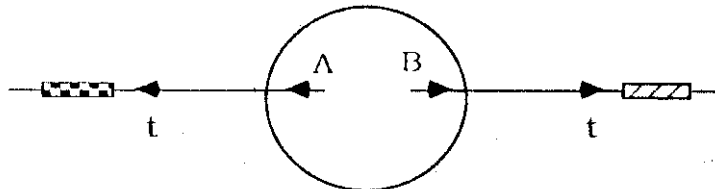


Fig. (4)

Consider a circular disc with some holes in it and place it on a smooth horizontal table. Get two spring balances, attach a hook to each string, then attach the two other ends of each string to two holes A and B (Say). Fix the ring of the balance attached to B (Say) to a fixed nail in the plane, then pull the other balance horizontally until the disc becomes at rest, we note that the two strings are on the same straight line and the readings in both balances are equal, (See Fig. 4). Clearly, the two forces have opposite directions.

We give some examples of equilibrium under the action of two

forces :

1. If a weight (w) is suspended by a light string, it balances under the action of two forces which are weight (w) and the tension in the string (t).

Therefore, $t = w$.

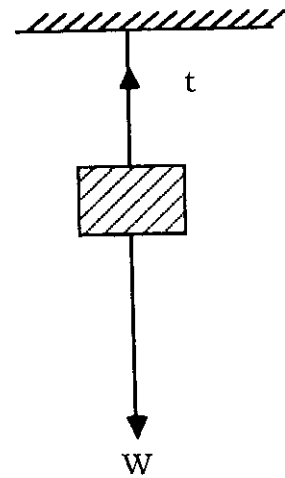


Fig. (5)

We notice that the string is vertical and the weight acts vertically downwards while the tension acts vertically upwards.

2. If a body whose weight is (w) is placed on a smooth horizontal plane, it balances under two forces which are (w) and the reaction of the horizontal smooth plane (r) (Fig. 6). The reaction of the plane is perpendicular to the plane vertically upwards.

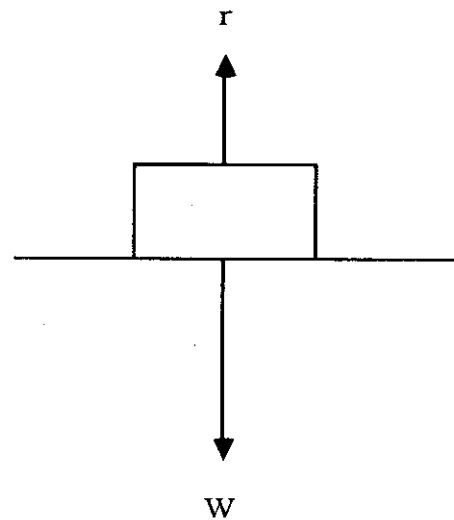


Fig. (6)

3. If two or more forces acted on a light rod at its two ends, then the rod is under the action of tension or pressure. (Fig. 7).



Fig. (7)

Two important rules :

1. If a rigid body is acted upon by two forces equal in magnitude, opposite in direction and their lines of action are on the same straight line, they have no effect on the body neither in case of rest or motion.
2. The exchange forces exerted from the action of a body upon another are always equal in magnitude and opposite in direction.

This is the third law of Newton which states :

"For every action there is a reaction equal in magnitude and opposite in direction"

It means that a body A acts on a body B by a force, the body B acts on A by another force equals to the first in magnitude and opposite in direction.

As a direct result of rule (1) :

If a force acts on a rigid body, we can transmit its point of action to any other point on the body and on the same line of action without change in its effect upon the body.

This result expresses the principle of (force's penetration) or the rule of (transmission of point of action).

If $\vec{F}_1$, $\vec{F}_2$ are two equal forces in magnitude and opposite in direction acting on a body at the points A, B, we can say that $\vec{F}_2$ is balanced with $\vec{F}_1$ regardless of its action on the line $\overleftrightarrow{AB}$ (i.e.) If $\vec{F}_2$ acts at A, it makes the same effect when it acts at B on the straight line. (Fig. 8).

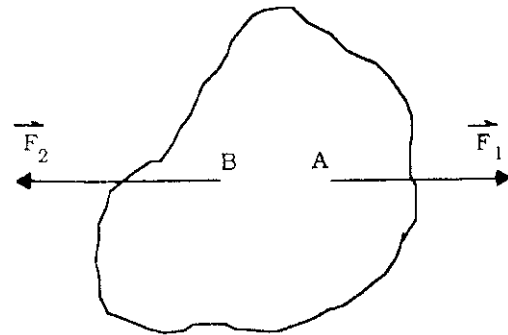


Fig. (8)

Remark :

We can use the same apparatus used in experiment (1) to prove the principle of (force's penetration), that is by using any hole on the extension of the line $\overleftrightarrow{AB}$.

Caution :

1. If a string passes over a smooth pulley, the tension in the string does not change in magnitude (Fig. 9-A), by passing by the pulley.

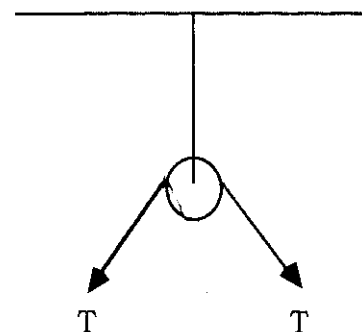
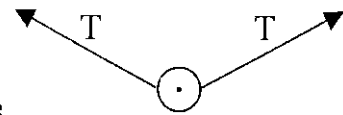


Fig. (9 A)

2. If a light string passes through a smooth ring, the tension in the string does not change. (Fig 9.B).



Equilibrium of three forces

Rule:

Fig. (9.B)

If three forces are acting at a point and can be represented by the sides of a triangle taken in the same cyclic order, then the forces are in equilibrium.

Proof:

Suppose that the vectors $\vec{OA}$, $\vec{OB}$ and $\vec{OC}$ represent completely the forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$, which meet at O respectively.

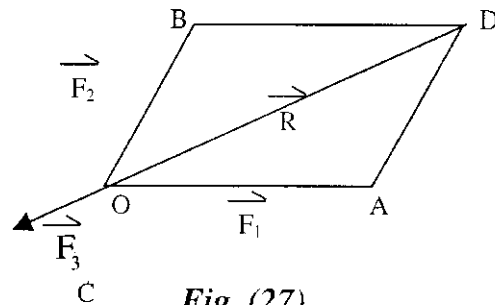


Fig. (27)

Complete the parallelogram OADB. Since $\vec{AD}$ equals and parallel to $\vec{OB}$, then $\vec{AD}$ represents $\vec{F}_2$.

Hence, if the three forces can be represented by the sides of triangle taken in the same cyclic order, the ΔOAD is that triangle, because $\vec{OA}$ represents $\vec{F}_1$, $\vec{AD}$ represents $\vec{F}_2$, then $\vec{DO}$ represents $\vec{F}_3$ but $\vec{OD}$ represents the resultant of $\vec{F}_1$ and $\vec{F}_2$.

$\vec{F}_3$ equals and opposite the resultant of $\vec{F}_1$ and $\vec{F}_2$ (i.e) $\vec{F}_3$ balances with the resultant of $\vec{F}_1$ and $\vec{F}_2$.

The forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$ are in equilibrium.

The triangle of Forces Rule

If three forces acting at a point be in equilibrium and a triangle is drawn whose sides are parallel to the lines of action of the forces, and taken in the same cyclic order then the lengths of the sides of the triangle are proportional to the magnitudes of the corresponding forces.

Proof :

By using the last figure, since the forces are in equilibrium, then $\vec{F}_3$ balances with the resultant of $\vec{F}_1$ and $\vec{F}_2$, therefore $\vec{DO}$ represents $\vec{F}_3$ while $\vec{OA}$ represents $\vec{F}_1$. $\vec{AD}$ represents $\vec{F}_2$, thus the sides $\vec{OA}$, $\vec{AD}$, $\vec{DO}$ of the ΔOAD are parallel to the directions of the forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$ respectively and their lengths are proportional to the magnitude of these forces (i.e.).

$$\frac{F_1}{OA} = \frac{F_2}{AD} = \frac{F_3}{DO}$$

Lami's Rule

If three forces meeting at a point and acting upon a particle are in equilibrium, then the magnitude of each force is proportional to sine of the angle between the other two forces.

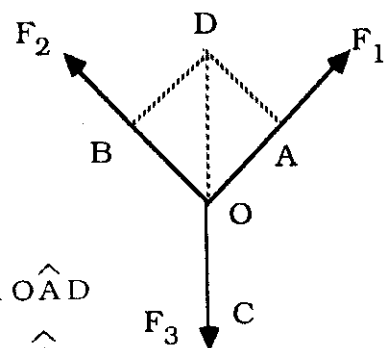
Proof :

Using the last figure :

$\therefore$ The forces are in equilibrium.

$$\begin{aligned} F_1 : F_2 : F_3 &= OA : AD : DO \\ &= \sin \widehat{ODA} : \sin \widehat{DOA} : \sin \widehat{OAD} \\ &= \sin \widehat{BOC} : \sin \widehat{COA} : \sin \widehat{BOA} \end{aligned}$$

by using the supplementing angles.



Example (1) :

A weight 100 gm. wt is suspended by two strings of lengths 30, 40 cm from two points on the same horizontal line with 50 cm apart. Find the tension in each string.

Solution :

By applying Lami's rule :

$$\frac{t_1}{\sin \theta_1} = \frac{t_2}{\sin \theta_2} = \frac{100}{\sin 90^\circ}$$

since AOB is right at O

$$\begin{aligned} \therefore t_1 &= 100 \sin \theta_1 \\ &= 100 \times \frac{30}{50} = 60 \text{ gm.wt} \\ t_2 &= 100 \sin \theta_2 \\ &= 100 \times \frac{40}{50} = 80 \text{ gm.wt} \end{aligned}$$

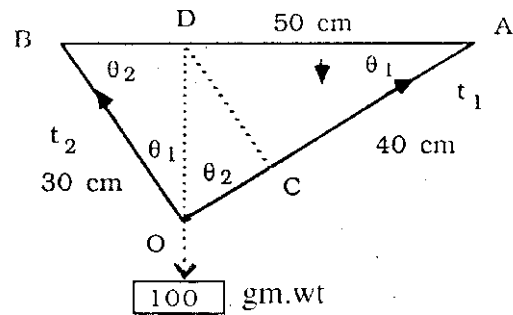


Fig. (28)

Another solution :

ODC is the triangle of forces where $\overline{OD}$ is vertical, $\overline{DC} \parallel \overline{OB}$

$$\therefore \frac{100}{DO} = \frac{t_1}{OC} = \frac{t_2}{CD}$$

$$\text{but } \frac{CD}{DO} = \sin \theta_2$$

$$\frac{OC}{DO} = \cos \theta_2$$

$$\therefore t_1 = 100 \times \frac{OC}{DO} = 100 \cos \theta_2 = 100 \times \frac{30}{50} = 60 \text{ gm.wt.}$$

$$t_2 = 100 \times \frac{CD}{OD} = 100 \sin \theta_2 = 100 \times \frac{40}{50} = 80 \text{ gm.wt.}$$

Example (2) :

A light string of length 24cm, its end A is fixed to a fixed point, a body of weight 100 gm.wt is hung at the other end B. Find the magnitude of the force required to keep the weight at a distance 12cm from the horizontal straight line through A in the following cases :

1. If the effective force is horizontal.
2. If the direction of the force is perpendicular to $\overline{AB}$. Find the tensions in the string in each case.

Solution :

- (1) In fig. (29), ABC could be considered the triangle of forces

$$\therefore \frac{100}{12} = \frac{t_1}{24} = \frac{F_1}{AC}$$

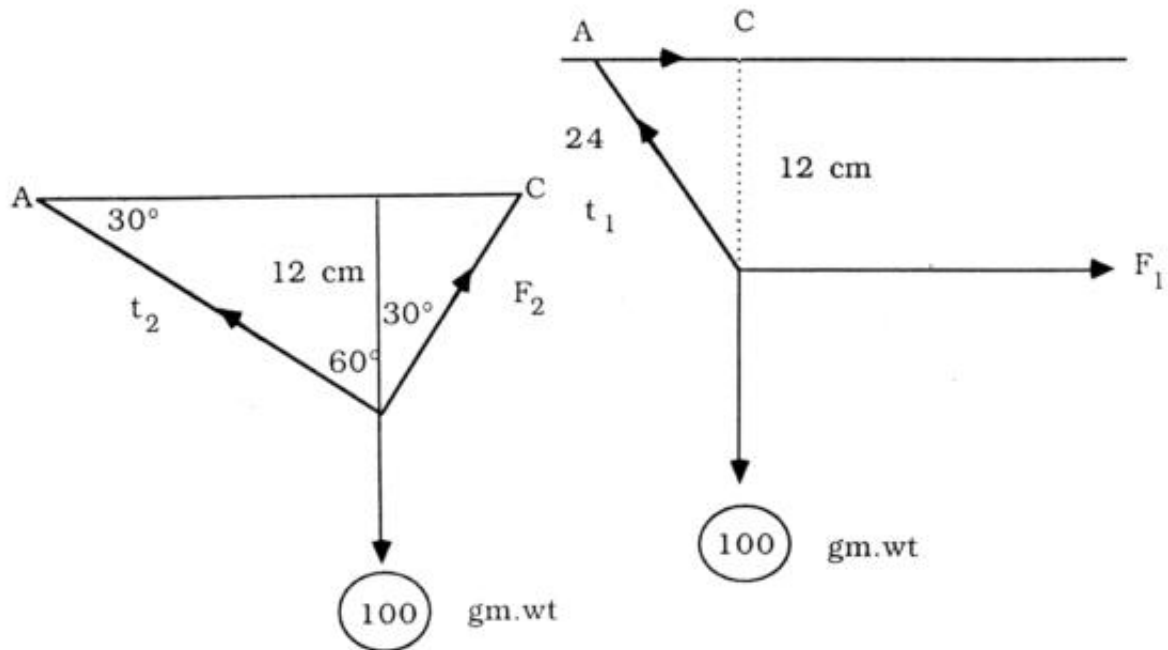


Fig. (29)

$$\begin{aligned} \text{Also } (AC)^2 &= (24)^2 - (12)^2 \\ &= 12 \times 36 \end{aligned}$$

$$\therefore AC = 12\sqrt{3} \text{ cm.}$$

$$\therefore \frac{F_1}{12\sqrt{3}} = \frac{100}{12}$$

$$\therefore F_1 = 100\sqrt{3} \text{ gm.wt.}$$

Similarly :

$$t_1 = \frac{100 \times 24}{12} = 200 \text{ gm.wt.}$$

(2) By applying Lami's rule :

$$\frac{100}{\sin 90^\circ} = \frac{t_2}{\sin 150^\circ} = \frac{F_2}{\sin 120^\circ}$$

$$t_2 = 100 \sin 150^\circ = 50 \text{ gm.wt.}$$

$$F_2 = 100 \sin 120^\circ = 50\sqrt{3} \text{ gm.wt.}$$

Example (3) :

A body of mass 6kg.wt is placed on a smooth inclined plane 30° with the horizontal, is supported by a force, find this force and the reaction of the plane in each of the following cases :

First : The force is horizontal.

Second : The force is inclined by 30° to the plane.

Solution :

First :

Applying Lami's rule :

$$\frac{F}{\sin 150^\circ} = \frac{r}{\frac{\sin 90^\circ}{6}} = \frac{6}{\sin 120^\circ}$$

$$\therefore \frac{F}{\frac{1}{2}} = \frac{r}{\frac{1}{2}} = \frac{6}{\frac{\sqrt{3}}{2}}$$

$$\therefore F = \frac{6}{\sqrt{3}} = 2\sqrt{3} \text{ kg.wt.}$$

$$r = 4\sqrt{3} \text{ kg.wt.}$$

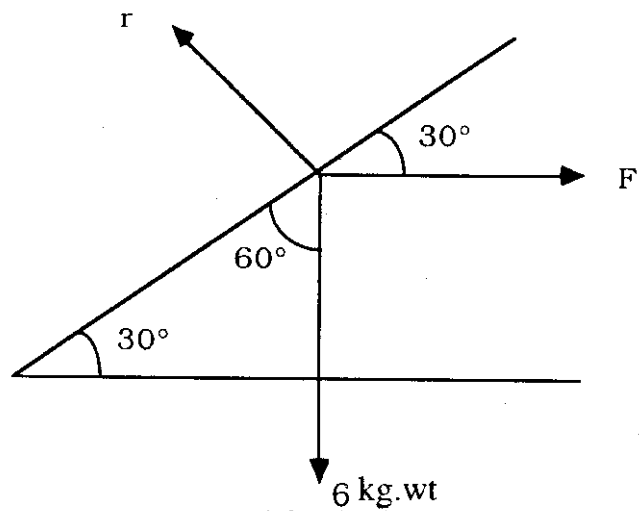


Fig. (30)

Second :

Applying Lami's rule

$$\frac{F}{\sin 150^\circ} = \frac{r}{\frac{\sin 150^\circ}{6}} = \frac{6}{\sin 60^\circ}$$

$$\therefore \frac{F}{\frac{1}{2}} = \frac{r}{\frac{1}{2}} = \frac{6}{\frac{\sqrt{3}}{2}}$$

$$\therefore F = r = 2\sqrt{3} \text{ kg.wt.}$$

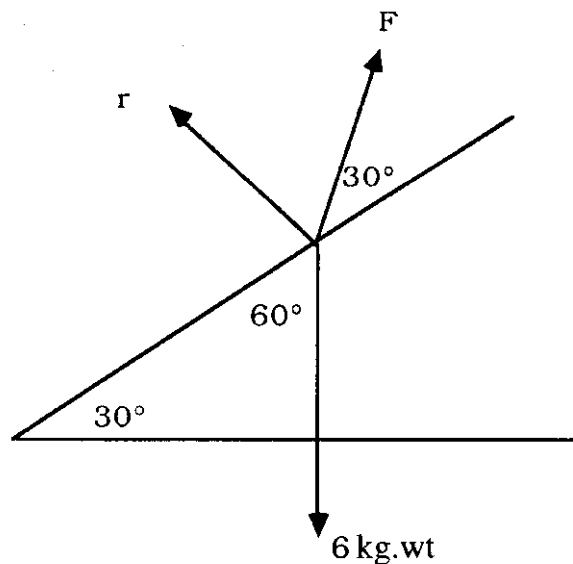


Fig. (31)

Exercise (4)

1. A body of weight 6.5 newtons is suspended by two strings of lengths 0.5 and 1.2 m. The two other ends are fixed at two points on a horizontal line such that the strings are perpendicular to each other. Find the tension.
2. Three coplanar forces 8, 10 and 12 newtons act at a particle. If the forces are in equilibrium, what is the measure of the angle between the other two forces ?
3. A body of weight "w" newtons is suspended by two strings, the first inclines by an angle of measure θ to the vertical passing over a smooth pulley and carries at its other end a body of weight 12 newtons, the other string forms with the vertical 30° , passing over a smooth pulley and carries at its other end a body of weight 8 newtons. Find w and θ .
4. A body of weight 60 newtons is placed on a smooth inclined plane to the horizontal by 30° . It is pulled up the plane by a string which coincides with the line of the greatest slope. Find the tension in the string and the reaction of the plane on the body.
5. The ball of a pendulum of weight 1 newton is displaced until the string makes 30° with the vertical under the action of a force perpendicular to the string. Find the magnitude of the force and the tension in the string.
6. A body of 400 gm.wt. is suspended at point A by a string, from a

point B on the string, another string is attached and pulled horizontally by a second string BC passing over a smooth fixed pulley and carries at its other end a body 300 gm.wt. Find the inclination of $\overline{AB}$ with the vertical and the tension in each of the two strings AB, BC.

7. A body of 72 gm.wt, is suspended at one end of a string, its other end of the string is fixed at a point A on a vertical wall. Another string is attached to the first one at a point B 25cm apart of A and pulled horizontally until the point B becomes 7cm, apart the wall. Find the tension in the horizontal string and in each part of the second string.

Equilibrium of a system of coplanar forces meeting at a point

Let the system of forces $\vec{F}_1, \vec{F}_2, \vec{F}_3$, which meet at a point and act in the same plane.

It is possible to determine the resultant $\vec{R}$ of these forces by drawing two perpendicular axes passing through the point of intersection of these forces and by resolving each force to two perpendicular components, in the directions of the axes.

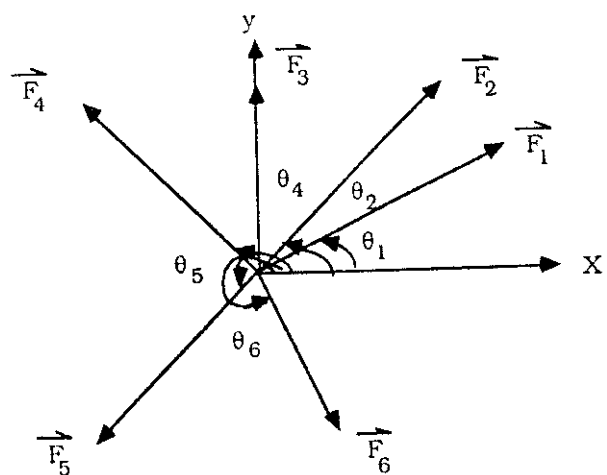


Fig. (32)

If X , Y are the algebraic sum of the components in the two directions $\vec{OX}$, $\vec{OY}$ respectively, then :

$$X = \sum_{r=1}^n F_r \cdot \cos \theta_r$$

$$Y = \sum_{r=1}^n F_r \cdot \sin \theta_r$$

where θ_r is the angle between the vector of the force $\vec{F}$ hence and $\vec{OX}$

$$R^2 = X^2 + Y^2$$

If this system of forces is in equilibrium then $R = 0$

$$\therefore X^2 + Y^2 = 0$$

This condition implies that :

$$X = 0 \text{ and } Y = 0$$

Since $\vec{OX}$, $\vec{OY}$ are two arbitrary perpendicular directions, then :

The condition of equilibrium of a system of forces meeting at a point is : The algebraic sum of the components in any two perpendicular directions must vanish.

Corollary :

The condition of equilibrium of many forces meeting at a point is that : (The algebraic sum of the components of these forces at any direction vanishes).

This result follows directly from the notion that the direction $\vec{OX}$ is one of the two directions of resolution which is arbitrary direction.

Remark :

We must notice that, it is not right to write down more than two independent equations, satisfied by the components of the forces. Therefore, we can solve the problem of equilibrium of several forces act at a point if the number of unknowns equals 2.

General rules of the equilibrium of a System of coplanar forces meeting at a point

The following rules could be applied in solving the problems of equilibrium of a body under the action of a number of coplanar forces meet at a point :

1. If a body is in equilibrium under the action of two forces, these two forces are equal in magnitude, opposite in direction and on the same line of action.
2. If the body is in equilibrium under the action of three coplanar forces such that two of them meet at a point, (non-parallel), the line of action of the third force must pass through this point.

To prove this rule, we notice that the resultant of the first two forces passes through their point of intersection, hence, we can consider that the body is in equilibrium under the action of two forces one of them is this resultant and the other is the third force, from the first rule, the third force and the resultant of the two forces must

be equal in magnitude, opposite in direction and their lines of action are on the same straight line, thus it passes through the point of their intersection.

These results help us in many cases to determine the line of action of the third force.

3. If the body is in equilibrium under the action of a number of coplaner forces meeting at a point, the conditions of equilibrium are :

The algebraic sum of the components in any direction must be vanished.

Example (1) :

From a point on the surface of a homogenous sphere, a light string is tied whose other end is attached to a point in a vertical smooth wall, if the sphere is stable on the wall, find the tension in the string and the reaction of the wall if the weight of the sphere is w newtons and acts at its centre and the string inclines to the vertical by 30° .

Solution :

In this problem we find that the sphere is in equilibrium state under the actions of three forces :

1. The weight (w) acts at M .
2. The reaction of the wall (R)
- $\therefore$ The wall is smooth, the reaction of the plane is perpendicular to the plane, therefore it passes through the centre of the sphere M .
3. The tension in the string (T).

Since the line of the action of the weight and the reaction of the wall passes through the centre M , then the line of action of the third force (the tension) must pass through the point M .

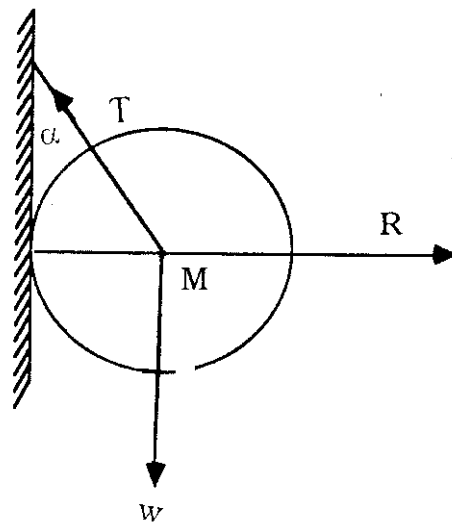


Fig. (33)

First : Algebraically :

By taking two perpendicular axes passing through the centre M one of them is horizontal and the other is vertical.

$$\therefore X = R - T \cos 60^\circ = 0 \dots \dots \dots (1)$$

$$Y = T \sin 60^\circ - w = 0 \dots \dots \dots (2)$$

$$\begin{aligned} \text{From (2) : } T \times \frac{\sqrt{3}}{2} &= w \\ T &= \frac{2w}{\sqrt{3}} \text{ newtons} \end{aligned}$$

$$= \frac{2\sqrt{3}}{3} w \text{ newtons}$$

By substitution in (1) :

$$R = 1/2 T = \frac{w}{\sqrt{3}} \text{ newtons} = \frac{w\sqrt{3}}{3} \text{ newtons}$$

Second : By applying Lami's rule :

$$\frac{w}{\sin 120^\circ} = \frac{T}{\sin 90^\circ} = \frac{R}{\sin 150^\circ}$$

$$\therefore T = \frac{w \sin 90^\circ}{\sin 120^\circ} = \frac{w \times 1}{\frac{\sqrt{3}}{2}} = \frac{2w}{\sqrt{3}} = \frac{2\sqrt{3}}{3} w \text{ newtons.}$$

$$R = \frac{w \sin 150^\circ}{\sin 120^\circ} = \frac{w \times \frac{1}{2}}{\frac{\sqrt{3}}{2}} = \frac{w}{\sqrt{3}} = \frac{\sqrt{3}}{3} w \text{ newtons.}$$

Third :

We can consider the triangle ABM as the triangle of forces because $\overrightarrow{MA}$ is the line of action of the force (t). $\overrightarrow{BM}$ is the line of action of the force (r), $\overrightarrow{AB}$ parallel to the line of force (w) and these directions are in the same cyclic order.

$$\therefore \frac{W}{AB} = \frac{R}{BM} = \frac{T}{MA}$$

$$\therefore R = \frac{w \cdot BM}{BM} = w \tan 30^\circ = \frac{w}{\sqrt{3}} = \frac{\sqrt{3}}{3} w \text{ newtons.}$$

$$T = \frac{w \cdot AM}{AB} = \frac{w \times 1}{\cos 30^\circ} = \frac{2w}{\sqrt{3}} = \frac{2\sqrt{3}}{3} w \text{ newtons.}$$

Example (2) :

A uniform rod of 1 metre long and 30 newtons weight is

suspended from both of its ends by two strings their other ends are fixed at one point in the ceiling. If the two strings are perpendicular, one of them is 60cm, find the tension in each string when it is suspended freely and in equilibrium (Consider the weight of the uniform rod acts at its midpoint).

Solution :

The rod is in equilibrium under the action of three forces :

1. Its weight 30 newtons acting at its midpoint M.
 2. The tension T_1 acts at $\overrightarrow{BO}$.
 3. The tension T_2 acts at $\overrightarrow{AO}$.
- $\therefore$ The two tensions meet at O then the weight must pass through O

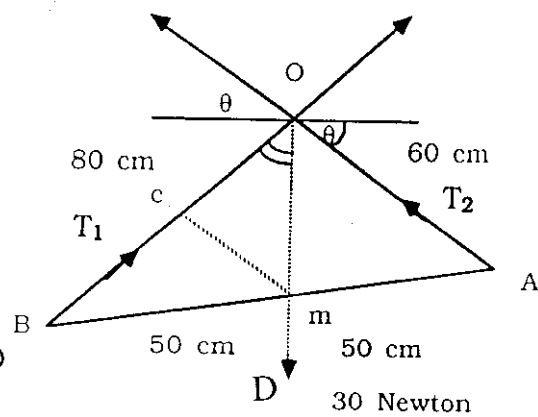


Fig. (34)

$\therefore$ ΔAOB is a right angled at O,

M is the midpoint of $\overline{AB}$

$$\therefore AM = MB = MO = 50 \text{ cm}, \tan \theta = \frac{60}{80} = \frac{3}{4}$$

$$\therefore \sin \theta = \frac{3}{5}, \cos \theta = \frac{4}{5}$$

By using Lami's rule :

$$\frac{30}{\sin 90^\circ} = \frac{T_1}{\sin (90^\circ + \theta)} = \frac{T_2}{\sin (180^\circ - \theta)}$$

$$T_1 = \frac{30 \cos \theta}{\sin 90^\circ} = 30 \cos \theta = \frac{30 \times 4}{5} = 24 \text{ newtons.}$$

$$T_2 = \frac{30 \sin \theta}{\sin 90^\circ} = 30 \sin \theta = \frac{30 \times 3}{5} = 18 \text{ newtons.}$$

Also triangle OMC is the triangle of forces :

$$\frac{30}{OM} = \frac{T_1}{CO} = \frac{T_2}{CM}$$

$$\therefore \frac{30}{50} = \frac{T_1}{40} = \frac{T_2}{30}$$

$$\therefore T_1 = \frac{30 \times 40}{50} = 24 \text{ newtons}$$

$$T_2 = \frac{30 \times 30}{50} = 18 \text{ newtons}$$

Exercise (5)

1. A smooth sphere of weight 15 newtons is on a smooth vertical wall and suspended by a light string from a point on its surface, the other end of the string is attached to a point on the wall above the point of contact between the wall and the sphere. If the length of the string equals the radius of the sphere, find the pressure on the wall and the tension in the string.
2. A sphere rests on two parallel rods lie in the same horizontal

plane, the distance between them equals the radius of the sphere. Find the pressure on each rod given that the weight of the sphere is 10 newtons.

3. A smooth iron sphere of weight 30 newtons rests between a vertical smooth wall and an inclined smooth plane inclines by 60° with the horizontal. Find the pressure on each of the wall and the plane.
4. A uniform rod which is movable around one of its ends is pulled aside by a horizontal force acting on the other end and equals half the weight of the rod. Find the angle of inclination of the rod to the vertical when it is in equilibrium and also the reaction at the first end.
5. A uniform rod of length 130cm and weight 26 newtons is suspended at its ends by two strings tied at one point, if the length of one of them is 50 cm and the length of the other one is 120cm, what is the position in which the rod is in equilibrium and what is the tension in each of the two strings.
6. AB is a uniform rod hinged at A, the other end B is tied to a string passes over a smooth pulley C exactly above A and attached to a weight equals half the weight of the rod. Find the angle of inclination of the rod to the horizontal in state of equilibrium given that $AC = AB$.

7. A uniform rod of weight 4 newtons is placed on two smooth planes inclined at 30° , 60° to the horizontal. Find the magnitude of the pressure on each plane and the angle of inclination of the rod to the horizontal in state of equilibrium.
8. A body is in equilibrium on a smooth inclined plane under the action of a force acting up the plane, its magnitude equals half the weight of the body. Find the angle of inclination of the plane to the horizontal and the reaction of the plane.

General Exercises

First: Complete the following :-

- (1) in $\triangle ABC$, $\overline{AB} + \overline{BC} + \overline{CA} = \dots\dots\dots$
- (2) In the quadrilateral ABCD, $\overline{BC} + \overline{CD} + \overline{DA} = \dots\dots\dots$
- (3) If $\vec{a} = 2\hat{i} - 3\hat{j}$, $\vec{b} = 3\hat{i} + \hat{j}$ then $\|\vec{a} - 2\vec{b}\| = \dots\dots\dots$
- (4) If $\vec{a} = 5\hat{i} + 4\hat{j}$, $\vec{b} = 5\hat{i} - 4\hat{j}$ then $\overline{AB} = \dots\dots\dots$
- (5) A body weighs 9 Newton is suspended at the end of string. It is in equilibrium under the effect of a horizontal force of magnitude F, when the string is inclined to the vertical by an angle whose measure is 30° then:

(a) $\frac{F}{\dots\dots\dots} = \frac{T}{\dots\dots\dots} = \frac{9}{\dots\dots\dots}$

(b) $F = \dots\dots\dots \text{ N} , T = \dots\dots\dots \text{ N}$

- (6) A body weighs w gm wt is attached at the End of a string whose length is 30 cm. The body is in equilibrium by the effect of a horizontal force of magnitude 36 gm wt, and is at distance 24 cm, then

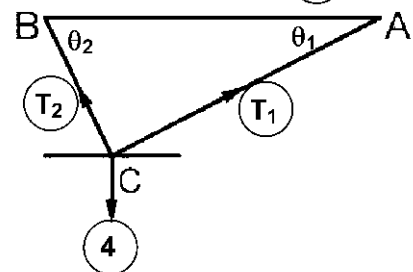
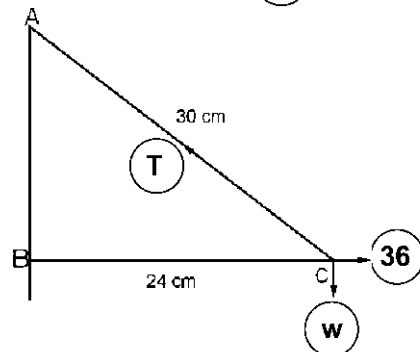
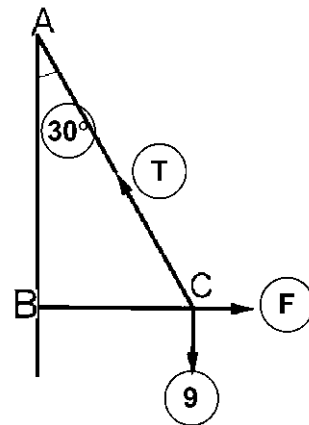
(a) $\frac{36}{\dots\dots\dots} = \frac{T}{\dots\dots\dots} = \frac{w}{\dots\dots\dots}$

(b) $T = \dots\dots\dots \text{ gm wt} ; w = \dots\dots\dots \text{ gm wt}$

- (7) A body weighs 4 Newton is suspended By means of two perpendicular strings Whose lengths are 6 cm, and 8 cm, the Other ends are fixed in two points A and B In the same horizontal level, then

(a) $\frac{T_1}{\dots\dots\dots} = \frac{T_2}{\dots\dots\dots} = \frac{\dots\dots\dots}{\sin 90^\circ}$

(b) $T_1 = \dots\dots\dots \text{ N} ; T_2 = \dots\dots\dots \text{ N}$



Second: Choose the correct answer from those between brackets :-

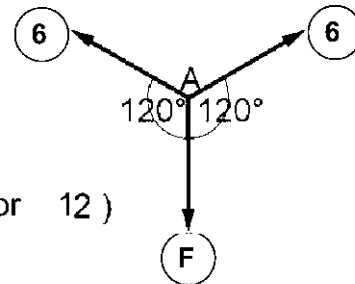
- (1) Two forces of magnitude 6N and 8N, if the magnitude of their resultant is 10N then the measure of the angle between the forces is
(0° or 60° or 90° or 180°)
- (2) Two forces of magnitude 4N and F N, the measure of the angle between them is 120° . If their resultant is perpendicular to the first force then $F = \dots\dots\dots$ N

(2 or 4 or $4\sqrt{3}$ or 8)

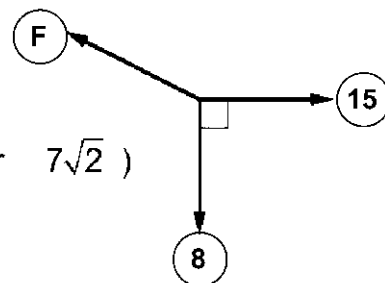
- (3) Two equal forces of magnitude 5 N. If the magnitude of their resultant is 5 N, then the angle between the forces measures °.
(0° or 90° or 120° or 180°)

- (4) If the resultant of two equal and perpendicular forces is 8 N, then the magnitude of each force N
($2\sqrt{2}$ or 4 or $4\sqrt{2}$ or 8)

- (5) Particle A is in equilibrium under the effect of
The three forces shown in the figure, where $\vec{F}$
Inclined with the two forces whose magnitudes
Are 6 and 6 N, by 120° then $F = \dots\dots\dots$ N
(6 or $6\sqrt{2}$ or $6\sqrt{3}$ or 12)



- (6) If the force $\vec{F}$ is in equilibrium with two
perpendicular forces of magnitudes
8 N and 15 N, then $F = \dots\dots\dots$ N
(7 or 17 or 23 or $7\sqrt{2}$)



- (7) If the three concurrent forces $\vec{F}_1 = 5\hat{i} - 3\hat{j}$, $\vec{F}_2 = 4\hat{i} + 6\hat{j}$, $\vec{F}_3 = k\hat{i} + n\hat{j}$, are in equilibrium then $k = \dots$, $n = \dots$
(9,3 or -9,-3 or -9,3 or 9,-3)

Third : Answer the following :-

- (1) $\triangle ABC$, $D, E \in \overline{BC}$ where $BD = CE$. Prove that $\overline{AB} + \overline{AC} = \overline{AD} + \overline{AE}$.
(2) $\triangle ABC$, $D \in \overline{BC}$ where $3BD = 4DC$. Prove that: $3\overline{AB} + 4\overline{AC} = 7\overline{AD}$.
(3) ABCD is a parallelogram its diagonals intersect at M. N is a point outside the figure. Prove that : $\overline{NA} + \overline{NB} + \overline{NC} + \overline{ND} = 4\overline{NM}$.
(4) ABCD is a trapezium where $\overline{AD} \parallel \overline{BC}$. E is the mid-point of $\overline{AB}$, F is the mid-point of $\overline{DC}$. Prove that: $\overline{AD} + \overline{BC} = 2\overline{EF}$
(5) ABCD is a quadrilateral, X is the mid-point of $\overline{AC}$, Y is the mid-point of $\overline{BD}$. Prove that : $\overline{AB} + \overline{AD} + \overline{CD} + \overline{CB} = 4\overline{XY}$.

(6) A force of magnitude 10 N. Find the components of this force in two directions, one makes 30° and the other makes 40° in the other direction of the force.

(7) Two forces of magnitudes F and $2F$, the measure of the angle between them is θ° . Find the value of θ if :

- a) The magnitude of their resultant is $3F$ N.
- b) The magnitude of their resultant is F N.

$$[\theta = 0^\circ ; \theta = 180^\circ]$$

(8) Two concurrent forces of magnitude F_1, F_2 N, the magnitude of their resultant is R Newton, where $4 \leq R \leq 16$ and $F_1 > F_2$. Find :

- a) The magnitudes of F_1, F_2
- b) The magnitude of R if $\theta = 60^\circ$.

$$[F_1 = 10 \text{ N} ; F_2 = 6 \text{ N} ; R = 14 \text{ N}]$$

(9) Two forces of magnitude 16 and F kg.wt. act on a particle, the measure of the angle between them is 120° . If their resultant is inclined to the force 16 kg.wt. by an angle whose measure is 30° . Find the magnitude of F and the resultant.

$$[F = 8 \text{ kg.wt.} ; R = 8\sqrt{3} \text{ kg.wt.}]$$

(10) Two forces of magnitude $2F$ and $3F$ N, the angle between them measures θ° . Find the value of θ if the magnitude of their resultant is :

- a) $3F$
- b) F
- c) $5F$
- d) $\sqrt{13}F$

$$[109^\circ 28' ; 180^\circ ; 0^\circ ; 90^\circ]$$

(11) Two forces act at a point, the value of one is 3 N more the other. If the magnitude of their resultant is $3\sqrt{3}$ N, and is perpendicular to the smaller force, then find the magnitude of each force and measure the angle between them.

$$[3, 5 \text{ N and } \theta = 120^\circ]$$

(12) Three coplanar forces of magnitude 5, 10, $4\sqrt{7}$ N, act on a particle. If the measure of the angle between the first and the second is 60° , then find the maximum and minimum values of the magnitude of the resultant of the three forces.

$$[9\sqrt{7} ; \sqrt{7}]$$

(13) Four coplanar forces of magnitude 2, 1, 4, $3\sqrt{3}$ N act at point A in directions of $\overrightarrow{BC}$; $\overrightarrow{BA}$; $\overrightarrow{CA}$ and $\overrightarrow{AD}$ where ABC is equilateral triangle and D is the mid-point of $\overline{BC}$. Find the magnitude and direction of their resultant.

$$[R = 1 \text{ N in direction of } \overrightarrow{AC}]$$

- (14) ABCDEF is a regular hexagon. The forces of magnitude 4, $8\sqrt{3}$, $4\sqrt{3}$, 8 kg.wt. act in the direction of $\overline{AB}$, $\overline{AC}$, $\overline{AE}$, $\overline{AF}$ respectively. Determine fully the resultant of these forces.

[R = 24 kg.wt. acts at A in direction of $\overline{AD}$]

- (15) ABCD is a square whose side is 12 cm long. E $\in$ BC where BE = 5 cm. Forces of magnitude 2, 13, $4\sqrt{2}$, 9 gm.wt. act in direction of $\overline{AB}$, $\overline{AE}$, $\overline{CA}$, $\overline{AD}$ respectively. Find fully the resultant of these forces.

[R = $10\sqrt{2}$ gm.wt. in direction of $\overline{AC}$]

- (16) ABCD is a rectangle in which AB = 6 cm and BC = 8 cm. Point E $\in$ BC where BE = 6 cm. The forces of magnitude 1, 10, $5\sqrt{2}$, 3 kg.wt. act in directions of $\overline{AD}$, $\overline{AC}$, $\overline{AE}$, $\overline{AB}$ respectively. Find the magnitude of their resultant and prove that its line of action passes through point E.

[R = $14\sqrt{2}$ kg.wt. makes an angle of measure 45° with $\overline{AD}$]

- (17) If $\vec{F}_1 = 5\hat{i} + 3\hat{j}$, $\vec{F}_2 = a\hat{i} + 6\hat{j}$, $\vec{F}_3 = -14\hat{i} + b\hat{j}$ are three coplanar forces act at a point, if their resultant is $R = (10\sqrt{2}, 135^\circ)$ then find the values of a and b.

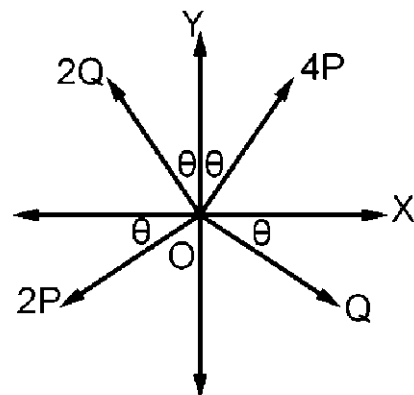
[a = -1 and b = 1]

- (18) The magnitudes of three coplanar concurrent forces are $F_1 = 8$, $F_2 = 4\sqrt{3}$ and $F_3 = 4$ gm.wt. If these forces are in equilibrium then find the measure of the angles between these forces.

[150° ; 90° , 120°]

- (19) In the figure opposite: Four coplanar concurrent forces act at the origin, in the directions shown in the figure. Where $\sin \theta = \frac{4}{5}$. If the resultant of these forces is $8\sqrt{2}$ N act in a direction makes 135° with $\overline{OX}$, then find the values of P and Q.

[P = 3 and Q = 14]



(20) Coplanar forces of magnitude F , $3\sqrt{2}$, $2\sqrt{3}$, $\sqrt{3}$ N act on a particle. The first force act in the East direction, the angle between the first and the second measures 45° , between the second and the third measures 105° , between the third and the fourth measures 120° . If the magnitude of their resultant is $3\sqrt{2}$ N, then find the value of F and measure the angle between the resultant and the first force.

(21) Five coplanar and concurrent forces act on a particle. The magnitudes of these forces are 12, 7, $5\sqrt{2}$, $7\sqrt{2}$, and 5 kg.wt. act in the directions East, North, North west, South west and South respectively. Find the magnitude and direction of the resultant, "describe the solution".

[$\vec{R} = \vec{0}$, ie $X=0$, $Y=0$, then the set of forces are in equilibrium]

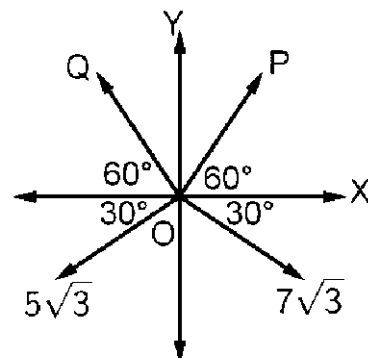
(22) ABCD is a trapezium right angled at C and $AD \parallel BC$, $BC = 18$ cm, $CD = 8$ cm, $AD = 12$ cm. The forces of magnitudes 10, P, Q kg.wt. act at point A in the directions of $\overline{AB}$, $\overline{CD}$, $\overline{BC}$ respectively. Find P and Q such that the set of forces is in equilibrium.

[$P = 8$ kg.wt. ; $Q = 6$ kg.wt.]

(23) In the figure opposite:-

Forces of magnitudes P, Q, $5\sqrt{3}$ and $7\sqrt{3}$ N act at O, and inclined to the X-axis by the angles whose measures are shown in the figure. If the set of forces is in equilibrium, then find the values of P and Q.

[$P = 3$ kg wt; $Q = 9$ kg wt]



(24) A body whose weight is 90 gm wt is placed on a smooth incline to the horizontal by an angle whose measure is 30° . The body is kept in equilibrium by means of a force. Find the magnitude of this force and the normal reaction of the incline if:

a) The force is horizontal.

b) The force is inclined to the horizontal by an angle of measure 60° .

[a) $30\sqrt{3}$, $60\sqrt{3}$ b) $F = N = 30\sqrt{3}$]

(25) A body of weight 6 N is placed on a smooth incline to the horizontal by an angle θ° . The body is kept in equilibrium by means of a force of magnitude $2\sqrt{3}$ N, inclined to the line of greatest slope of the plane by an angle of measure θ° up, find the value of θ and the magnitude of the normal reaction of the plane.

[$\theta = 30^\circ$; $R = 2\sqrt{3}$ N]

(26) In the figure opposite :-

A body of weight 9 kg wt is placed on
A smooth incline to the horizontal by 30° .

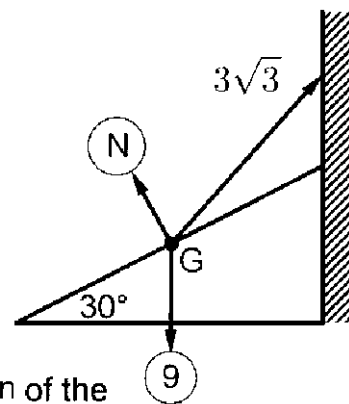
The body is kept in equilibrium by

Means of a tension force $T = 3\sqrt{3}$ kg wt

Acts along a string one end of which is

Fixed by the body and the other end at
A point on a vertical wall at the top of the

Incline. Find the measure of the angle of inclination of the
string to the horizontal and the magnitude of the reaction of the plane on the
body.



[60° , $3\sqrt{3}$ kg wt]

(27) A weight of 200 gm wt is suspended at the end of a light string, the other end of which is attached to the ceiling of a room. The weight is pulled by a horizontal force until the string is inclined to the vertical by an angle of measure 30° . Find the magnitude of the horizontal force and the tension in the string.

[$\frac{400\sqrt{3}}{3}$, $\frac{200\sqrt{3}}{3}$]

(28) A weight of 200 gm wt is suspended by means of two light inextensible strings, one of them is inclined to the vertical by an angle of measure θ° and the other inclined to the vertical by an angle of measure 30° . If the magnitude of the tension in the first string is 100 gm wt, then find θ and the magnitude of the tension in the second string.

[60° , $100\sqrt{3}$ gm wt]

(29) AB is a uniform rod which is 40 cm long, and weighs 12 N. The rod rests with its end A on a vertical smooth wall. It is kept in equilibrium by means of a light inextensible string one of its ends is attached to point C where $C \in$

AB where $BC = 10$ cm, and the other end is fixed to point D on the wall directly vertical above A. If the rod is then inclined by an angle whose measure is 60° to the vertical, then find the magnitudes of the tension in the string and the reaction of the wall.

[$8\sqrt{3}$, $4\sqrt{3}$ N]

(30) AB is a uniform rod, its end A is attached by a smooth hinge fixed in a vertical wall. A horizontal force is then act at the end B, to keep the rod in equilibrium while it is inclined to the wall by an angle of measure 45° . If the weight of the rod is 4 kg wt, acts at its mid-point, then find the magnitude of the force and the reaction of the hinge.

[2 , $2\sqrt{5}$ kg wt]

II - D Y N A M I C S

CHAPTER I

THE RELATIVE VELOCITY

(1-1) Velocity vector

When we say that a car moves with a velocity of 60 km/h we mean that this car covers a distance of 60 km in one hour if it moves with this velocity all the time. Sometimes it is not sufficient to give the magnitude of the velocity, but it is also necessary to determine its direction.

(Remark : The magnitude of the velocity vector is called speed)

Definition : The velocity vector of a particle is the vector of which the magnitude is equal to the value of the velocity (speed) and its direction is the same as the direction of the motion.

Units of speed : km/h, m/sec, cm/sec.

Uniform motion is the case where the magnitude and the direction of the velocity vector are constant.

Here we have two important remarks on the uniform motion

- a The direction of the velocity vector is constant means that the particle moves along a constant direction.
- b The magnitude of the velocity vector is constant means that the particle covers equal distances in equal times

The relation between displacement and velocity vector in the case of the uniform motion

$$\vec{S} = t\vec{V} \text{ or } \vec{V} = \frac{1}{t} \vec{S}$$

The average velocity vector - the instantaneous velocity vector

The average velocity vector of a particle between two successive moments t_1 , t_2 and devoted by the symbol $\vec{V}_a$, is defined as follows displacement vector between this two moments divided by the time interval $(t_2 - t_1)$.

For example : If we notice a particle moving from the position O and let the particle be at A_1 and A_2 after two successive moments t_1 and t_2 as shown in fig.(1), respectively. If $\vec{r}_1$ and $\vec{r}_2$ position vectors of the particle at the moments t_1 and t_2 then

$$\vec{S} = \vec{r}_2 - \vec{r}_1$$

$$\vec{V}_a = \frac{\vec{S}}{t_2 - t_1} = \frac{\vec{r}_2 - \vec{r}_1}{t_2 - t_1}$$

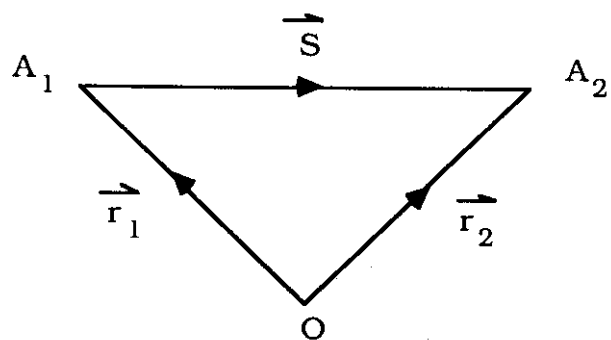


Fig. (1)

Let the time interval $(t_2 - t_1)$ be very small and time t the middle of this interval $(t_2 - t_1)$. In this case the average velocity vector is

called the instantaneous velocity vector of the particle at the moment t and denoted by the symbol $\vec{V}$.

Variable motion

If the motion of a particle is not uniform we call it variable motion. In the case of variable motion the velocity vector changes from time to time either in magnitude or direction or both.

Example :

A cyclist cut 20 km on a straight road with a velocity 18 km/hr, then he returns and cut 12 km in the opposite direction at 12 km/hr. Find his average velocity direction during the whole journey.

Solution :

Let the cyclist started his motion at A and moved to B then from B to C.

Let $\hat{e}$ is a unit vector has the same direction of $\vec{AB}$. (Fig. 2)

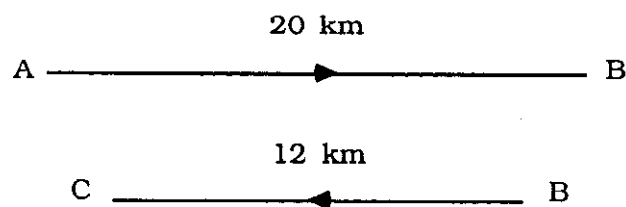


Fig. (2)

$$\text{Time taken from A to B} = \frac{20}{18} = 1.11 \text{ h.}$$

$$\text{Time taken from B to C} = \frac{12}{12} = 1 \text{ h.}$$

$$\text{Total time} = 1.11 + 1 = 2.11 \text{ h.}$$

$$\begin{aligned}
 \text{Total displacement} \quad \vec{S} &= \vec{AB} + \vec{BC} = \vec{AC} \\
 &= 20\hat{e} - 12\hat{e} = 8\hat{e} \\
 \therefore \vec{V}_a &= \frac{8\hat{e}}{2.11} = 3.8\hat{e}
 \end{aligned}$$

The $\vec{V}_a$ vector has the same direction of $\hat{e}$ and its norm is about 3.8 km/h. It is important to know that the norm of the total displacement is 8 km but the whole distance is 32 km.

THE RELATIVE VELOCITY

The experiment shows that motion is something relative to another and it changes in description by the change of observer. Each of us imagines that the lighting poles seem to be moving when we see it from the windows of a slowly moving train. In the same time if you look from the train window to the moving cars in the same direction of the motion of the train, they seem to move very slowly while we feel the contrary if the cars are moving in the opposite direction of the train. All these phenomenas emphasize the fact of relativity of the motion.

We mentioned that the motion seems to be different on observing it by certain person who is walking or in a car or in an aeroplane for example, the student knows well that we can observe the motion of a rocket by an aeroplane or the motion of clouds and

storms by means of a satellite or the motion of ships by other ones. We will concentrate on one property only of the motion properties, it is (The vector of velocity of a moving particle).

Consider two particles : A, B we suppose that their motion is observed by a certain person on the earth. Fig. (3) let $\vec{V}_a, \vec{V}_b$ be the velocity vector of the particles A, B.

Respectively w.r.t. to a person (O). What will be the case if certain person existing on the particle (A) and moving with it to watch the motion of the particle (B). This person will find out that the particle B is moving at certain velocity, we give it the symbol V_{ba} and we call it the velocity of (B) relative to (A).

Then what is the relation between

the vectors $\vec{V}_a, \vec{V}_b, \vec{V}_{ba}$

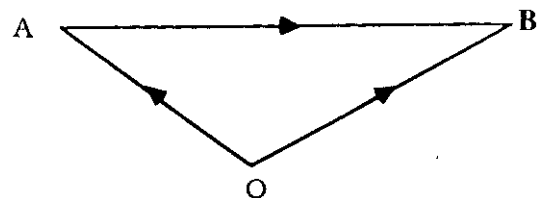


Fig. (3)

To answer this question, let us think as follows :

To consider that the particle (A) is observing the motion of particle (B) if the particle (A) is at rest like the observer at point (O). So we give particle (A) an extra velocity. $\vec{V}_a$ hence it will move by

the resultant of the two velocities $\vec{V}_a$, $-\vec{V}_a$ which means that its resultant velocity is the zero vector, and so we could stop (A.) So as not to change the mechanical state of the group consisting of the 2 particles A, B. We must give the particle (B) a similar extra velocity, i.e., $(-\vec{V}_a)$ then its resultant velocity will be $(\vec{V}_b - \vec{V}_a)$. Fig. (4).

So the observer on particle (A) who moves with it will observe the motion of particle (B) with a velocity $(\vec{V}_b - \vec{V}_a)$, and this the velocity of (B)w. r. t. (A).

We get now the important relation which can find a definition to the velocity vector (B) relative to (A).

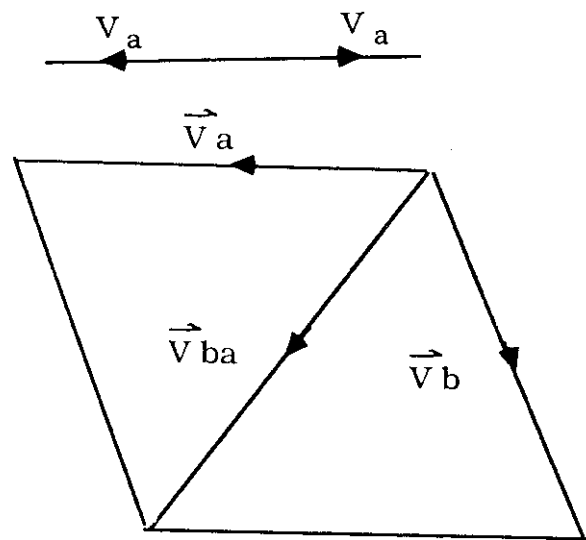


Fig. (4)

$$\vec{V}_{ba} = \vec{V}_b - \vec{V}_a \dots\dots\dots (1)$$

and it is a relation which gives the relative velocity when knowing the two velocities of the two particles w.r.t. the observer at point (O), we can write this relation as follows :

$$\vec{V}_b = \vec{V}_a + \vec{V}_{ba} \dots\dots\dots (2)$$

By this relation we can calculate $\vec{V}_b$ knowing $\vec{V}_a$, $\vec{V}_{ba}$.

To illustrate the importance of the relation No. (2), we suppose

that (O) is a control station for satellites, and that particle (A) is a satellite while particle (B) is a star which cannot be directly watched from the control station, then we let the satellite watch it, so we get $\vec{V}_{ba}$ also the velocity of the satellite will be known to the station i.e. $\vec{V}_a$ is also known, from relation (A) we can easily calculate $\vec{V}_b$.

Example :

A cyclist (A) moves on a straight with velocity 12 km/h is he meets another cyclist (B) moving with 10 km/h. Find the velocity of (B) relative to (A) in case of (B) moving in an opposite direction to the direction of (A) and in case of moving in the same direction as (A).

Solution (1) :

Let $\hat{C}$ be the unit vector having the same direction as the velocity vector (A).

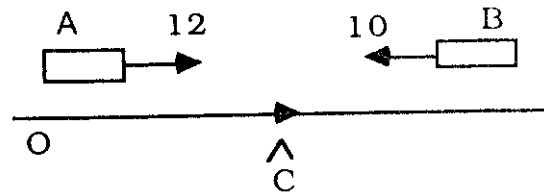
First :

Cyclist (B) moving in opposite direction of the motion of (A). Fig. (5).

$$\vec{V}_a = 12 \hat{C}$$

$$\vec{V}_b = 10 \hat{C}$$

$$\therefore \vec{V}_{ba} = -10 \hat{C} - 12 \hat{C} = -22 \hat{C}$$

**Fig (5)**

i.e., The cyclist (A) will feel that the cyclist (B) is moving towards him at a velocity of 22 km per hour.

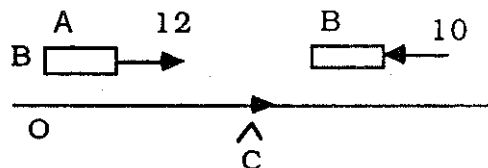
Second :

Cyclist (B) is moving in the same direction as (A) Fig. (6)

$$\vec{V}_a = 12 \hat{C}$$

$$\vec{V}_b = 10 \hat{C}$$

$$\therefore \vec{V}_{ba} = 10 \hat{C} - 12 \hat{C} = -2 \hat{C}$$

**Fig. (6)**

i.e., The cyclist (A) will feel that the cyclist (B) is moving towards him (i.e., backwards) at a velocity of 2 km/h.

Exercise (1)

- 1) A cyclist covered 40 km on a straight way with a velocity of 20 km/h. Then he returned back with a velocity of 15 km/h. Find the average velocity during his whole trip.
- 2) A train covered the distance between Cairo and Alexandri in two stages.

1st stage : From Cairo to Tanta the covered distance was 100 km, and the velocity was 100 km/h.

2nd stage: From Tanta to Alexandria the covered distance was 110 km and the velocity was 80 km/h.

The stopping of the train in Tanta was 10 minutes. Find the average velocity of the train during its whole trip.

(Consider that the train was moving the whole distance in a straight line)

- 3) A controlling speed car (Radar) moves on the desert road (Caro - Alex) at 30 km/h. It watched a truck coming from the other opposite road which seemed to be moving at 110 km/h. Find the real velocity of the truck.
- 4) A steamer was moving in a straight way towards a port. At 50 km from it, and aeroplane passed over it in the opposite direction and at 250 km/h, it observed the steamer which seemed to the

aeroplane as it is moving with 275 km/h. Find the time elapsed from the moment the aeroplane observed the steamer till it reaches the port ?

- 5) A car (A) moving on a straight way measured the relative velocity of another car (B) coming from the opposite direction it was 140 km/h. When the car (A) reduced its velocity to its half and remeasured the relative velocity of car (B) it found it 120 km/h Find the actual velocities of the two cars ?
- 6) Two aeroplanes fly at the same speed in a straight line while one of them is following the other with a distance 500 m apart. At a certain moment the one behind launched a rocket at the leading one. This rocket reached the first one after 2 seconds of its launching. What is the velocity of the rocket ?
- 7) A war ship and a steamer are moving in a straight line at the same velocity, every one of them was moving towards the other, the war ship watching the motion of the steamer it seemed to it that it is moving at 80 km/h. When the distance between them became 6 km, the war ship launched a torpedo on the steamer if the engine of the torpedo can push it at 100 km/h. What is the time taken from the moment of launching and the moment the torpedo reaches the steamer ?

CHAPTER II

Variable motion - The straight motion with uniform acceleration

(2-1) Variable motion

If a particle moves such that its velocity vector changes from an instant to another and this change is either in magnitude or direction or both, then we say that this particle makes a variable motion or is moving with acceleration.

Definition of the acceleration vector $\vec{A}$

It is the rate of change of the velocity vector with respect to the time or it is the change of the velocity vector in the unit of time.

For example : If we consider two successive moments t_1 and t_2 and let $\vec{V}_1$ and $\vec{V}_2$, be the velocity vectors of the particle at t_1 and t_2 respectively, then

$$\vec{A} = \frac{\vec{V}_2 - \vec{V}_1}{t_2 - t_1}$$

Units of measurement of the magnitude of the acceleration vector

$$\text{Norm of the acceleration vector} = \frac{\text{Norm of the velocity vector}}{\text{time}}$$

$$\text{Unit of acceleration} = \frac{\text{Unit of velocity}}{\text{unit of time}}$$

$$(\text{cm/sec}) / \text{sec i.e. cm/sec}^2$$

$$\text{or } (\text{m/sec}) / \text{sec i.e. m/sec}^2$$

In some cases the unit of time changes and then the measure unit of the norm of the acceleration vector changes too. For example :
(Km/h)/sec.

We must be careful when we change the unit of the norm of the acceleration vector. This is shown in the following examples :

$$(1 \text{ km/h})/\text{sec} = \frac{1000\text{m}}{3600 \text{ sec} \times \text{sec}} = \frac{5}{18} \text{ m/sec}^2.$$

$$(1 \text{ km/h})/\text{sec} = \frac{1000 \times 100 \text{ cm}}{3600 \text{ sec} \times \text{sec}} = \frac{250}{9} \text{ cm/sec}^2.$$

(2-2) Uniform variable motion :

In general the acceleration vector of a particle changes in magnitude or direction or both from a moment to another. But there are cases where the acceleration vector is constant all the time. Then we say that the motion of the particle is a uniform variable motion or the particle is moving with a uniform acceleration.

Now we will study the laws which describe this motion in the special case where the particle moves on a fixed straight line in space. It is sufficient to have the velocity vector at a certain moment parallel to the constant acceleration vector. Knowing that we have

studied these laws in first year secondary of physics, we have to use the following symbols :

$\vec{U}$: to denote the velocity vector of the particle at the beginning of the measuring time.

$\vec{V}$: to denote the velocity vector at the end of a time interval

$\vec{S}$: to denote the displacement vector of the particle during the time interval t .

$\vec{A}$: to denote the acceleration vector.

First law : The relation between velocity and time in case of linear motion with uniform acceleration.

$$\vec{A} = \frac{\vec{V} - \vec{U}}{t} \qquad \vec{V} = \vec{U} + \vec{A} t$$

Let a unit vector $\hat{C}$ be parallel to the fixed straight line on which the particle moves. Then the relation is

$$V \hat{C} = U \hat{C} + (A \hat{C}) t$$

and we can write this relation as follows ;

$$\mathbf{V} = \mathbf{U} + \mathbf{A}t$$

We know that the algebraic measure of any of these vectors is :

- a) positive if the direction of the vector is the same as the direction of $\hat{C}$
- b) negative if the direction of the vector is opposite to the direction of $\hat{C}$

Second law : The relation between displacement and acceleration in case of linear motion with uniform acceleration.

$S = \text{average velocity} \times \text{time interval } t$

$$S = V_a \times t = (u + \frac{1}{2} At)t$$

$$S = Ut + \frac{1}{2} At^2$$

Third law :

The relation between displacement and velocity at linear motion with uniform acceleration.

$$S = u \left(\frac{V - U}{A} \right) + \frac{1}{2} A \left(\frac{V - U}{A} \right)^2$$

$$2 AS = V^2 - U^2$$

$$V^2 = u^2 + 2 AS$$

Example 1 : A particle started its motion in a constant direction with initial velocity 10 cm/sec and with uniform acceleration 4 cm/sec² in the same direction of the initial velocity.

Find : a) its velocity at the end of one minute.

b) the displacement covered during this minute.

Solution :

$$V = u + At \quad \therefore \quad V = 10 + 4 \times 60 = 250 \text{ cm/sec}$$

$$s = ut + \frac{1}{2} At^2 \quad \therefore \quad S = 10 \times 60 + \frac{1}{2} \times 4 \times 3600$$

$$= 600 + 7200$$

$$= 7800 \text{ cm}$$

Example 2 :

The velocity of a car uniformly decreased from 66 Km/h to 12 km/h after it covered 585 m. Find the time taken by the car to cover this displacement and the distance which will be covered afterwards to come to rest.

Solution :

$$U = 66 \times \frac{5}{18} = \frac{55}{3} \text{ m/sec} \quad \text{and } V = 12 \times \frac{5}{18} = \frac{10}{3} \text{ m/sec}$$

$$\therefore V^2 = U^2 + 2As$$

$$\therefore V = U + At$$

$$\therefore \left(\frac{10}{3}\right)^2 = \left(\frac{55}{3}\right)^2 + 2A \times 585 \quad \therefore \frac{10}{3} = \frac{55}{3} - 5t$$

$$A = \frac{-5}{18} \text{ m/sec}^2$$

$$\therefore t = 54 \text{ sec}$$

$$\therefore V^2 = U^2 + 2As$$

$$\therefore 0 = \left(\frac{10}{3}\right)^2 - 2 \times \frac{5}{18} \times S$$

$$\therefore S = 20 \text{ m}$$

Example 3 :

A particle starts its motion in a constant direction at 10 cm/sec and uniform acceleration of 4 cm/sec² in the same direction of its velocity.

Calculate :

1st : the distance covered by the particle in the 5th second only.

2nd : the distance covered by the particle in the 8th and 9th seconds together

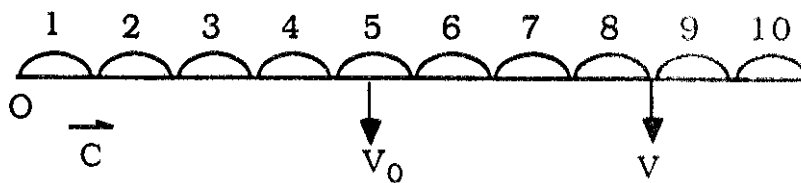


Fig. (7)

We consider that the direction of motion is the positive direction. Fig. 7.

$$\therefore U = 10 \text{ cm/sec.}$$

$$\& A = 4 \text{ cm/sec.}$$

1st : To find the distance covered by the particle at the 5th second only. We find the total distance it covered during the 1st five seconds, and the total distance covered by the particle during the 1st four seconds.

$\therefore$ The difference between the two distances is the required distance covered during the 5th sec, only.

$$\therefore S = Ut + \frac{1}{2} At^2$$

$$\therefore S_5 = 10 \times 5 + \frac{1}{2} \times 4 \times 25 = 100 \text{ cm.}$$

$$\& S_4 = 10 \times 4 + \frac{1}{2} \times 4 \times 16 = 72 \text{ cm.}$$

As the motion is in the positive direction so the magnitude of the displacement vector is equal to the covered distance.

∴ The distance covered at the 5th sec only.

$$= 100 - 72$$

$$= 28 \text{ cm.}$$

Another Solution :

We find the average velocity during the 5th sec = the actual velocity at the middle of the time interval.

(i.e) The velocity after $4 \frac{1}{2}$ sec.

$$\therefore V = U + At$$

$$\therefore V = 10 + 4 \left(4 \frac{1}{2}\right) = 28 \text{ cm/sec.}$$

$$\therefore S = \text{the average velocity} \times \text{time.}$$

$$S = 28 \times 1 = 28 \text{ cm.}$$

Which equals the distance covered during the 5th second only.

(Second) : By using the same method we can find the distance covered during the 8th and 9th sec, that is by finding S_7 , S_9 the difference between them is the required distance.

Another Solution :

We find the average velocity for the 8th and 9th sec, which equals the velocity at the mid-point during this interval, equals the velocity after 8 sec of starting

$$\begin{aligned}\therefore V &= U + At \\ &= 10 + 4 \times 8 = 42 \text{ cm/sec.}\end{aligned}$$

$\therefore$ the distance = Average velocity $\times$ time

$\therefore$ the required distance = $42 \times 2 = 84 \text{ cm.}$

Example 4 :

A body moves with uniform acceleration in a straight line. If it covered 26 m during the 4th second of its motion, 56 m during the 9th. Calculate the acceleration by which it moves, and its initial velocity.

Solution :

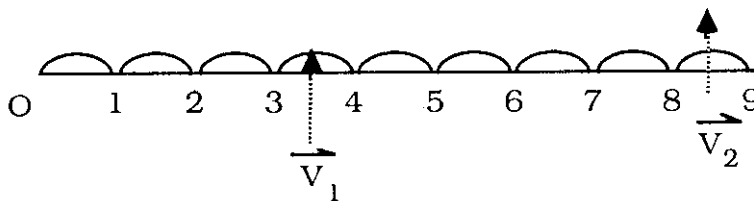


Fig. (8)

Let us consider the (+ve) direction is the direction of motion.

$$\begin{aligned}\therefore \text{ the average velocity} &= \frac{\text{displacement}}{\text{time}} \\ &= \frac{\text{distance}}{\text{time}}\end{aligned}$$

$$\therefore V_1 = \frac{26}{1} = 26 \text{ m/sec which is the velocity at :}$$

$$t_1 = 3 \frac{1}{2} \text{ sec}$$

$$V_2 = \frac{56}{1} = 56 \text{ m/sec which is the velocity at :}$$

$$A = \frac{V_2 - V_1}{t_2 - t_1} = \frac{56 - 26}{8\frac{1}{2} - 3\frac{1}{2}}$$

$$\therefore V = U + At$$

$$V_1 = U + At_1$$

$$\therefore 26 = U + 6 \times 3\frac{1}{2}$$

$$\therefore U = 5 \text{ cm/sec.}$$

Which is the velocity at the beginning of motion.

Exercise (2)

- 1) A particle moved in a constant direction, it covered 18 m during the 1st three seconds of its motion, 12 m during the 5th second, 20 m in the 9th second. Prove that the particle was moving with uniform acceleration. Calculate its velocity at the beginning of motion.
- 2) A, B, C and D are four points on the same straight line such that $AB = BC = 2640$ cm. A body is moving from point A with uniform acceleration, it covered the distance AB in 40 sec, BC in 60 sec. In how many seconds it covers the distance $\overline{CD}$, what is the length of $\overline{CD}$ if the body came to rest at D ?
- 3) A train moves between two stations of 700 m apart with acceleration of $1 \frac{1}{2}$ m/sec². It started its motion from rest at the 1st station for 10 seconds, then with uniform velocity for sometimes, then it covered the last 60 m, with uniform retardation until it came to rest at the next station. Find the time taken to cover the distance between the two stations.
- 4) A train started its motion from a station A with uniform acceleration 15 cm/sec². It reached its maximum velocity after 2 minutes and moved with this velocity for $11 \frac{2}{15}$ minutes, afterwards the brakes are used and it continued its motion

with a uniform retardation 150 cm/sec^2 until it came to rest at B. Find the distance AB and its average velocity to cut this distance.

- 5) A car started its motion from rest with uniform acceleration 180 km/h for every minute. After 24 seconds the acceleration is ceased and the velocity decreased uniformly by means of friction and air resistance at a rate of 450 m/h/sec and after 32 seconds the brakes are used to stop it in 8 seconds. Find the distance covered by the car.
- 6) A car is moving with uniform velocity 72 km/h passed by a police car at rest, the police car started motion following it after 10 seconds with uniform acceleration a distance 100 m until the velocity became 90 km/h , then it moved with this velocity until it overtook the first car. Find the time taken during this process from the beginning of motion of the police car and the distance covered by this car.
- 7) A particle is moving with uniform acceleration, it covered 200 m during the first 4 sec, then 50 m during the 7th and 8th sec. Find its initial velocity and the distance covered from the beginning of its motion until it came to rest.
- 8) A bullet is fired horizontally towards a wooden target with a velocity 200 m/sec , it was penetrated 4 cm. Find the uniform

acceleration, with which the bullet moved. If the target is 3 cm thick, what is the velocity by which the bullet comes out ?

- 9) The velocity of a car at a certain instant is 5 m/sec, it was moving with uniform acceleration until it reached its maximum velocity after 20 sec, it moved with this velocity for 28 sec, then by uniform retardation for 12 sec until it came to rest. If the total distance is 1150 m, find each of the two accelerations.
- 10) A particle was moving with uniform acceleration (A). It covered 400 cm in 10 sec, then it increased the acceleration to ($2A$), so it covered another 700 cm in 10 sec, and its velocity at the end of this distance was 90 cm / sec, then it moved with retardation ($3A$) until it came to rest. Evaluate (A) and the total distance covered by the particle.

CHAPTER III

VERTICAL MOTION UNDER GRAVITY

The acceleration by which bodies fall freely (i.e., the air resistance is neglected) is called "The acceleration of gravity." It is always acting towards the center of earth and denoted by $\vec{g}$

This acceleration has a constant magnitude for all the bodies in the same place whatever their weights or their volumes or their chemical instruction, it differs by the change of latitude line, so it decreases at equatorial and increases when we get closer to the poles.

It also differs according to the height above the sea level, it decreases when we get higher and increases when we approach it.

Its value at the sea level at the equatorials 978.024 cm/sec^2 and at the Northern pole 983.210 cm/sec^2 . In London it is 981 cm/sec^2 . In general we are going to consider the gravity 980 cm/sec^2 or 9.8 m/sec^2 to facilitate the calculations.

LAWS OF VERTICAL MOTION :

Considering the acceleration of gravity is constant, we can apply the previous rules of motion on a straight line with uniform acceleration after replacing the symbol "A" by "g" to denote the gravity, then the rules will be in the following forms :

$$V = U + gt \dots\dots\dots (1)$$

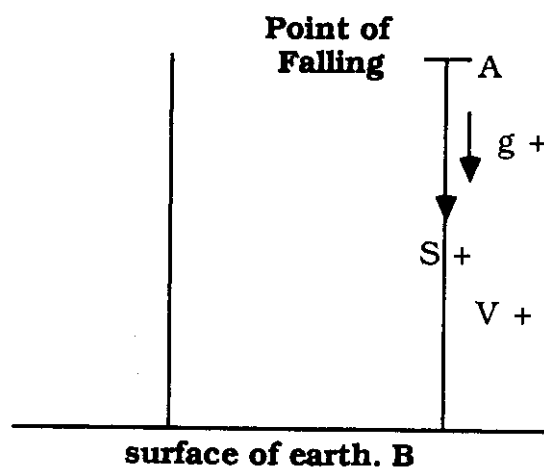
$$S = Ut + \frac{1}{2} gt^2 \dots\dots\dots (2)$$

$$V^2 = U^2 + 2gS \dots\dots\dots (3)$$

Notice that these relations are between the algebraic measurements of $\vec{V}$, $\vec{U}$, $\vec{g}$ and $\vec{S}$, when we apply these rules we have to notice the signs of each of the acceleration of gravity g and the velocities V, U and the displacement S and follow :

FIRST : IF THE BODY IS FALLING OR PROJECTED DOWNWARDS :

In this case it is better to consider the vertical direction downwards is the positive direction, since the gravity always acts towards the centre of earth (i.e.) vertically downwards then the acceleration of gravity ($+g$). since the displacement S is measured from the point of falling downwards it is also, ($+S$) and the velocity of the particle downward is also ($+V$) Fig. (9).



SECOND : IF THE BODY IS PROJECTED VERTICALLY UPWARDS :

In this case we take the upward vertical direction is a positive direction, since the acceleration of gravity is always acting vertically downwards, then it is negative, since the initial velocity U in this case is vertically upwards then it is positive. Fig. (10)

But the signs of U and S depends on the instant of measure as it will be clear in the following example :

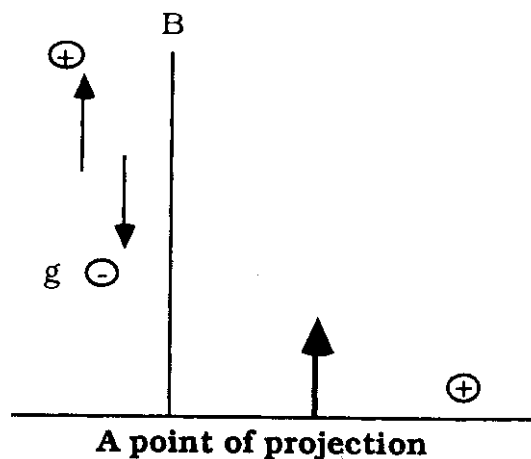


Fig (10)

Remark :

We can write the rules of vertical motion under the action of gravity in case of projecting upward in the following form :

$$V = U - gt \quad (1)$$

$$S = Ut - \frac{1}{2} gt^2 \quad (2)$$

$$V^2 = U^2 - 2 gS \quad (3)$$

In which we consider the vertical direction upward is positive and $g = \|\vec{g}\|$

THE TIME AND THE DISTANCE OF THE MAXIMUM HEIGHT :

If a body is projected vertically upwards, and it considered the vertical upwards direction is positive, then the velocity of projection will be positive and the acceleration of gravity will be negative, therefore it decreases the velocity of the body until it comes to zero. It is said that the particle reached its maximum height above the point of projection.

Then, the body changes its direction descending, so we take the vertical downward as a direction, the velocity and the acceleration are positive too, then it increases the velocity of the body until it reaches the point of projection again.

1. To find the time taken to reach the maximum height :

$$\therefore V = U - gt \text{ (because the body is projected upward)}$$

$$\therefore 0 = U - gt$$

$$\therefore gt = U$$

$$\therefore t = \frac{U}{g}$$

$$\text{(i.e.) The time of maximum height} = \frac{\text{Velocity of projection}}{\text{The acceleration of gravity}}$$

2. To find the maximum height :

$$\therefore V^2 = U^2 - 2 gS \text{ (because the body is projected upward)}$$

$$\therefore 0 = U^2 - 2 gS$$

$$\therefore 2 gS = U^2$$

$$\therefore S = \frac{U^2}{2g}$$

$$\text{Maximum height} = \frac{\text{The square of the velocity of projection}}{\text{Twice the acceleration of gravity}}$$

Rule

If a body is projected vertically upwards, then :

1. time taken to reach the maximum height equals time of moving downwards.
2. the velocity, of projection from the point of projection upwards is equal to the velocity with which the body reaches the point of projection.

The Proof

First :

$$\therefore S = Ut - \frac{1}{2} gt^2$$

Replace S by Zero

$$\therefore 0 = Ut - \frac{1}{2}gt^2$$

$$gt^2 - 2Ut = 0$$

$$\therefore t(gt - 2U) = 0$$

$$\therefore t = 0 \quad \text{or} \quad \frac{2U}{g}$$

then $t = 0$ is the time of projecting, $t = \frac{2U}{g}$ is the total time from the instant of projection until it returns to the point of projection.

$$\therefore \text{time of arrival the maximum height} = \frac{2U}{g}$$

$$\therefore \text{time of descending} = \frac{2U}{g} - \frac{U}{g} = \frac{U}{g}$$

$$\therefore \text{time of ascending} = \text{time of descending.}$$

Second :

$$\therefore V^2 = U^2 - 2gS$$

replace S by Zero

$$\therefore V^2 = U^2$$

$$\therefore V = \pm U$$

$$\therefore +U \text{ is the velocity of projecting upward}$$

$$-U \text{ is the velocity of projecting downward.}$$

Example 1 :

A stone is projected in a well with velocity 2m/sec, it reached its bottom after 3 sec, find the velocity of the stone which it strikes the base of the well and find the depth of the well.

Solution :

First :

To get the velocity :

$$V = U + gt$$

$$\begin{aligned}\therefore V &= 2 + 9.8 \times 3 \\ &= 2 + 29.4 = 31.4 \text{ m/sec.}\end{aligned}$$

Second :

To calculate the distance :

$$\begin{aligned}\therefore S &= Ut + \frac{1}{2} gt^2 \\ \therefore S &= 2 \times 3 + \frac{1}{2} \times 9.8 \times 9 \\ &= 6 + 4.9 \times 9 \\ &= 6 + 44.1 = 50.1 \text{ m}\end{aligned}$$

which is the depth of the well.

Example 2 :

A particle is projected vertically upward with velocity 28 m/sec.

Find the time elapsed to reach the maximum height and the distance attained.

Solution :

$$\therefore V = U - gt \qquad 0 = 28 - 9.8 t$$

$$\therefore t = \frac{28}{9.8} = \frac{280}{98} = \frac{20}{7} = 2\frac{6}{7} \text{ sec}$$

$$\therefore V^2 = U^2 - 2gS$$

$$\therefore 0 = (28)^2 - 2 \times 9.8 S$$

$$\therefore S = \frac{28 \times 28}{2 \times 9.8} = 40 \text{ m.}$$

Another Solution :

$$\text{Time of maximum height} = \frac{U}{g} = \frac{28}{9.8} = \frac{280}{98} = \frac{20}{7} = 2\frac{6}{7} \text{ sec}$$

$$\text{The maximum height} = \frac{U^2}{2g} = \frac{(28)^2}{2 \times 9.8} = 40 \text{ m.}$$

Example 3 :

A body is projected vertically with a velocity of 14.7 m/sec. from a point on the ground. Calculate the velocity of the body when it becomes at a height 9.8 m from the ground. Explain the meaning of the two answers.

Solution :

Let us take the upward vertical direction a positive direction Fig. (11).

$$\therefore U = + 14.7 \text{ m/sec.}$$

$$\therefore S = 9.8 \text{ m.}$$

$$\therefore V^2 = U^2 - 2gS$$

$$\therefore V^2 = (14.7)^2 - 2 \times 9.8 \times 9.8$$

$$= (4.9)^2 (9-8) = (4.9)^2$$

$$\therefore V = \pm 4.9 \text{ m/sec.}$$

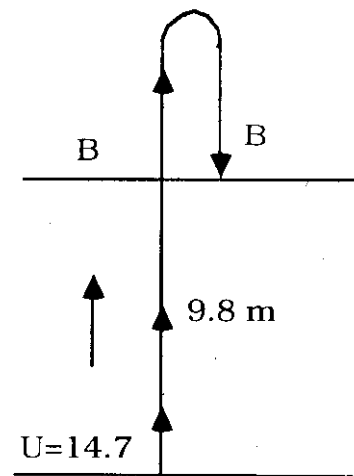


Fig. (11)

$V = + 4.9$ m/sec. when the body is moving upwards.

$U = - 4.9$ m/sec. when the body is moving downwards.

(i.e.) The body will be at height 9.8 m from the ground twice, the first when it was moving upward with velocity 4.9 m/sec as Fig. (11) the second when it was moving downward with the same velocity at B' which is on the same level of B.

Remark

If a body is projected vertically upwards, it moves on the vertical line passing through the point of projection and returns on the same line, but to illustrate the problem we draw the line of descending beside that of ascending.

Example 4 :

A particle is projected vertically upward with velocity 24 m/sec. Find the time taken until the body reaches a point 32.4 m below the point of projection.

Solution :

Let us take the upward direction is the (+ve) direction.

$$\therefore U = + 24 \text{ m/sec.}$$

$$\therefore S = -32.4 \text{ m.}$$

$$\therefore S = Ut - \frac{1}{2} gt^2$$

$$\therefore - 32.4 = 24t - \frac{1}{2} \times 9.8t^2$$

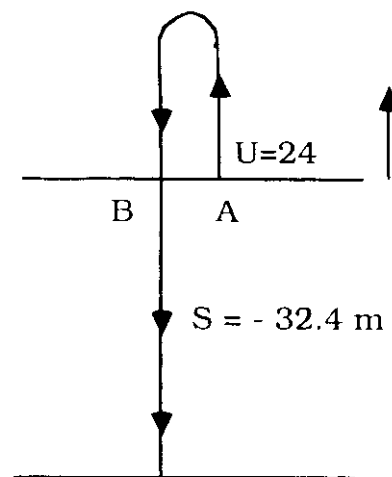


Fig. (12)

$$\therefore 4.9 t^2 - 24 t - 32.4 = 0$$

$$\therefore 4.9 t^2 - 240 t - 374 = 0$$

$$\therefore (t - 6) (49 t + 54) = 0$$

$$\therefore t = 6 \text{ or } t = -\frac{54}{49} \text{ (refused)}$$

$\therefore$ the time is 6 sec.

Exercise (3)

- 1) A body fell vertically downward. Find its velocity after 4 sec and the time elapsed to cover 68.6 m.
- 2) A stone is projected in a well with velocity 4 m/sec. It reached its bottom after 2 sec. Find the depth of the well and the stone velocity when it strikes its bottom.
- 3) A particle is projected vertically upward with velocity 19.6 m/sec. Find the time taken to reach the maximum height and this maximum height.
- 4) A particle is projected vertically upward with velocity 24.5 m/sec, after how many second it comes back the point of projection ?
- 5) A particle moves vertically with velocity 19.6 m/sec from a point on the ground. Find when it becomes at 14.7 m above the ground and find its velocity then. Explain the meaning of the two answers.

- 6) A particle is projected upward with velocity 14 m/sec . Find the time taken to reach a point 350 m below the point of projection.
- 7) A stone is projected vertically upward with velocity 19.6 m/sec from the top of a tower 156.8 m height. Find when it reaches the ground and its velocity at this instant.
- 8) A small ball is projected vertically upward from the window of a house, the ball was observed moving downwards against the window after 4 sec then reached the ground after 5 sec from the instant of projection. Find the height of the window in metres.

CHAPTER 4

DIFFERENTIATION OF VECTOR FUNCTIONS

In the Mechanics course, the student became familiar with many of the motion characteristics, and knew that some of these, like position, velocity, acceleration are described as vectors.

When motion occurs, these vectors may change in magnitude or direction or both from instant to instant, i.e, it forms functions of an independent variable, the time.

These functions are called “Vector Functions”

When studying the motion, we need sometimes to find the changes that may happen in the motion characteristics of the vector functions, or precisely, to calculate the time rate of change of these functions. The student may remember that, by the mathematical expression about the rate of change of a function we mean its derivative with respect to the independent variable, and thus we are going to give the definition of the vector function, taking the position vector function $\mathbf{r}(t)$ as a special example to our study, but still the results are true for any other vector functions.



Consider the motion of a particle, let A, B be the positions of this particle at two successive and very close instants $t, t + h$ respectively as in fig. (13).

Take a fixed origin O, to which the motion is related, then

$$\vec{r}(t) = \overrightarrow{OA}, \vec{r}(t+h) = \overrightarrow{OB}$$

The displacement vector is

$$\overrightarrow{AB} = \vec{r}(t+h) - \vec{r}(t).$$

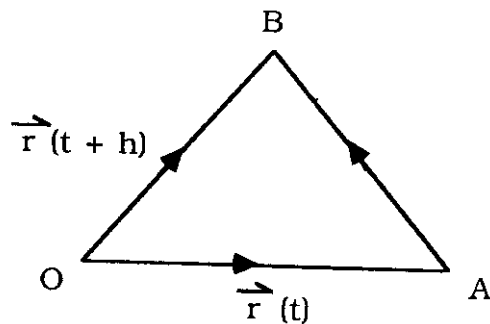


Fig. (13)

Dividing this vector by the value of the time interval h , we obtain the new vector $\frac{\vec{r}(t+h) - \vec{r}(t)}{h}$

Now we calculate the limit of this vector when the time interval h tends to zero if it exists, it is called "The derivative of the vector function $\vec{r}(t)$ with respect to the variable t ".

Definition :

The derivative of the vector function $\vec{r}(t)$ with respect to the independent variable t , denoted by $\frac{d\vec{r}}{dt}$ is defined from the relation

$$\frac{d \vec{r}}{dt} = \lim_{h \rightarrow 0} \frac{\vec{r}(t+h) - \vec{r}(t)}{h} \quad (1)$$

Since the independent variable here is the time, the derivative $\frac{d \vec{r}}{dt}$ expresses the time of change to the vector function $\vec{r}(t)$.

Also since the limit of a vector function is a vector, then “The derivative of a vector function is also a vector”.

If the vector $\vec{r}$ is a constant vector, then $\frac{d \vec{r}}{dt} = \vec{0}$, i.e. The derivative of a constant vector is the zero vector.

As we are concerned only with the applications of the derivatives of vector functions on the rectilinear motion of a particle, we are going to consider the special case in which the vector function $\vec{r}(t)$ is parallel to a fixed direction, defined by the unit vector $\hat{C}$, and thus we can express the vector function $\vec{r}(t)$ in terms of its algebraic measure, as follows.

$$\vec{r}(t) = r(t) \hat{C}$$

The derivative or $\vec{r}(t)$ is obtained as follows :

$$\frac{d \vec{r}}{dt} = \lim_{h \rightarrow 0} \frac{r(t+h) \hat{C} - r(t) \hat{C}}{h}$$

But $\hat{C}$ is a constant vector

$$\frac{d \vec{r}}{dt} = \lim_{h \rightarrow 0} \left[\frac{r(t+h) - r(t)}{h} \right] \hat{C}$$

The quantity between the square brackets is equal to the derivative of the algebraic measure $r(t)$ at the instant t .

$$\therefore \frac{d \vec{r}}{dt} = \frac{d r}{dt} \hat{C} \quad (2)$$

It is clear from the last relation that the derivative $\frac{d \vec{r}}{dt}$ is parallel to the function $\hat{C}$.

Example (1)

If $\vec{r}(t) = 3 t^2 \hat{C}$, where t is the time, $\hat{C}$ is a constant unit vector, find $\frac{d \vec{r}}{dt}$ and determine its algebraic measure at $t = 3$.

Solution :

$$\frac{d \vec{r}}{dt} = \frac{d}{dt} (3 t^2) \hat{C} = 6 t \hat{C}$$

$\therefore$ the algebraic measure of the derivative

$$\frac{d \vec{r}}{dt} \text{ equals } 6 t$$

At $t = 3$, the algebraic measure of the derivative

$$= 6 \times 3 = 18.$$

Example (2)

Find the derivative of the vector function

$$\vec{r}(t) = (t^2 - 4 t + 1) \hat{C}$$

where t is the time, $\hat{C}$ is a fixed unit vector. Find its algebraic measure, and its magnitude at any instant t , and at $t = 1$

Solution :

$$\frac{d \vec{r}}{dt} = \frac{d}{dt} (t^2 - 4 t + 1) \hat{C}$$

$$\frac{d \vec{r}}{dt} = (2t - 4) \hat{C}$$

$\therefore$ the algebraic measure of the derivative = $2t - 4$

The magnitude of the derivative =

$$\|(2t - 4) \hat{c}\| = |2t - 4| \|\hat{c}\|$$

At $t = 1$: The algebraic measure = $2 \times 1 - 4 = -2$

$$\text{The magnitude} = |-2| = 2$$

Velocity vector as a derivative of the position vector :

In what follows we will be concerned only with the rectilinear motion of a particle i.e. the motion of a particle in a straight line which is fixed in space. Let $\hat{c}$ be a unit

vector parallel to this line, O is the origin on this line. Let A be the position of the particle at time, t ,

B its position at a subsequent instant $(t + h)$ very near to t .

$\vec{r}(t) = \vec{OA}$, $\vec{r}(t + h) = \vec{OB}$ are

the position vectors of the particle

at these two instants, $\vec{s} = \vec{r}(t + h) - \vec{r}(t)$

is the displacement vector during this interval of time defined by

these instants, fig. (14).

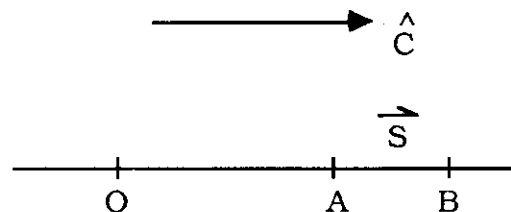


Fig. (14)

The average velocity vector $\vec{v}_{av} = \frac{\vec{s}}{h} = \frac{\vec{r}(t+h) - \vec{r}(t)}{h}$

If we let h very close to zero, $\vec{v}_{av}$ tends to the instantaneous velocity vector of the particle at the instant t . Therefore by taking the limit of both sides of the above relation when h tends to zero, we get:

$$\lim_{h \rightarrow 0} \vec{v}_{av} = \lim_{h \rightarrow 0} \frac{\vec{r}(t+h) - \vec{r}(t)}{h}$$

$$\text{or } \vec{v} = \frac{d\vec{r}}{dt} \quad (3)$$

which means that "The velocity vector of a particle at an instant is equal to the derivative of its position vector at this instant".

Also since the motion is in a straight line, the vector $\vec{r}$ is parallel to the vector $\vec{v}$, and we can express each by its algebraic measure relative to the unit vector $\hat{c}$ as follows :

$$\vec{r} = r(t) \hat{c}, \quad \vec{v} = v(t) \hat{c}$$

and equation (3) takes the form

$$v \hat{c} = \frac{d}{dt} (r \hat{c}) = \frac{dr}{dt} \hat{c}.$$

eliminating $\hat{c}$ from both sides

$$v = \frac{dr}{dt} \dots\dots\dots (4)$$

i.e. The algebraic measure of the velocity vector of the particle at a certain instant is equal to the derivative of the algebraic measure of its position vector at this instant.

Also we can formulate the two relations (3), (4) in terms of the displacement instead of the position vector as follows :

$\vec{r}_0$ be the position vector at the starting instant i.e. at $t = 0$ say, and that $\vec{s} = \vec{r} - \vec{r}_0$ is the displacement vector of the particle in the interval between the starting instant and time t .

$$\vec{r}_0 \text{ is a constant vector} \quad \therefore \frac{d\vec{r}_0}{dt} = \vec{0}$$

$$\therefore \frac{d\vec{r}}{dt} = \frac{d\vec{s}}{dt}$$

$$\text{Substituting in (3) we get} \quad \vec{v} = \frac{d\vec{s}}{dt} \quad (5)$$

i.e. The instantaneous velocity vector of a particle at a certain instant is equal to the derivative of its displacement vector between the initial instant and this instant.

Also if s is the algebraic measure of the displacement vector $\vec{s}$ relative to the unit $\hat{c}$, i.e.

$$\vec{s} = s \hat{c}$$

relation (5) takes the following form as a relation between the algebraic measures of the two vectors $\vec{v}$ and $\vec{s}$:

$$\boxed{v = \frac{ds}{dt}} \quad (6)$$

i.e.

The algebraic measure of the velocity vector at any instant is equal to the derivative of the algebraic measure of the displacement vector of the particle between the initial instant and this instant.

Acceleration vector as a derivative of the velocity vector:

We notice that the average acceleration vector $\vec{a}_{av}$ of the particle during the interval of time t to $(t + h)$ is given as follows :

$$\vec{a}_{av} = \frac{\vec{v}(t + h) - \vec{v}(t)}{h}$$

and when $h \rightarrow 0$ $\vec{a}_{av}$ tends to the instantaneous acceleration vector of the particle at the time instant t .

Therefore by taking the limit of both sides of the last relation as $h \rightarrow 0$ we obtain :

$$\lim_{h \rightarrow 0} \vec{a}_{av} = \lim_{h \rightarrow 0} \frac{\vec{v}(t + h) - \vec{v}(t)}{h}$$

$$\therefore \vec{a} = \frac{d\vec{v}}{dt} \quad (7)$$

which means that :

"The instantaneous acceleration of the particle at any instant is equal to the derivative of its velocity vector at this instant.

Writing $\vec{v} = v \hat{c}$, we find that

$$\vec{a} = \frac{d}{dt} (v \hat{c}) = \frac{dv}{dt} \hat{c}$$

This shows that the acceleration vector $\vec{a}$ is also parallel to the unit $\hat{c}$, and if a is the algebraic measure of the acceleration vector $\hat{a}$ i.e. if $\vec{a} = a \hat{c}$ then relation (7) takes the following form

$$a \hat{c} = \frac{dv}{dt} \hat{c}$$

$$\boxed{\therefore a = \frac{dv}{dt}} \quad (8)$$

i.e.

The algebraic measure of the acceleration vector of a particle at any instant is equal to the derivative of the algebraic measure of its velocity vector at this instant.

Example (3) :

A particle moves so that its position vector $\vec{r}$ is given as a function of the time t in terms of a constant unit vector $\hat{c}$ as follows :

$$\vec{r} = (t^2 - 3t) \hat{c}.$$

Find the velocity and acceleration vectors of the particle at any time instant. Find also the magnitude of its velocity at the starting instant $t = 0$ and prove that the motion is a retardation at this instant.

Solution :

$$\begin{aligned} \text{Instantaneous velocity vector } \vec{v} &= \frac{d\vec{r}}{dt} \\ &= (2t - 3) \hat{c} \end{aligned}$$

At the starting instant $t = 0$, we get

$$\vec{v}_0 = -3 \hat{c}, \text{ i.e. } \vec{v}_0 \text{ is in an opposite direction to that of } \hat{c}.$$

Also the magnitude of its initial velocity = 3 units of velocity

Instantaneous acceleration vector :

$$\vec{a} = \frac{d\vec{v}}{dt} = 2 \hat{c} \text{ which is a constant vector}$$

i.e. acceleration vector $\vec{a}$ is in the same direction as $\hat{c}$. It is clear that $\vec{v}_0$, $\vec{a}$ are in opposite directions, which means that the motion is a retardation at $t = 0$.

Example (4)

If the position vector of a particle is given by the relation

$$\vec{r} = (t^3 - 2t^2 + 2t) \hat{c}.$$

Find the velocity and acceleration vectors, and prove that the motion was a retardation at $t = 0$, and accelerated at $t = 1$.

solution :

$$\vec{v} = \frac{d\vec{r}}{dt} = (3t^2 - 4t + 2) \hat{c}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = (6t - 4) \hat{c}$$

at $t = 0$ we have

$$\vec{v} = \vec{v}_0 = 2 \hat{c}, \quad \vec{a} = \vec{a}_0 = -4 \hat{c}$$

$\therefore$ the motion is a retardation at $t = 0$.

at $t = 1$

$$\vec{v} = \vec{v}_1 = \hat{c}, \quad \vec{a} = \vec{a}_1 = 2 \hat{c}$$

$\therefore$ the motion is accelerated at $t = 1$.

Example (5) :

If the position of a particle is given as a function of the time in the form $\vec{r} = (2t^2 + 2) \hat{c}$. Find the displacement vector at the instant t , and hence find the velocity vector of the particle.

Solution :

Position vector at the instant $t = 0$

$$\vec{r}_0 = 2 \hat{c}$$

$$\begin{aligned}
 &= \vec{r} - \vec{r}_0 = (2t^2 + 2) \hat{c} - 2 \hat{c} \\
 &= 2t^2 \hat{c} \\
 \vec{v} &= \frac{d\vec{s}}{dt} = 4t \hat{c}
 \end{aligned}$$

To find if the motion is a retardation or an acceleration we have to find first the acceleration vector.

$$\vec{a} = \frac{d\vec{v}}{dt} = 4 \hat{c}$$

We notice that the vector $\vec{v}$ is always in the same direction as vector $\vec{a}$ at any time instant $t > 0$.

$\therefore$ the motion is accelerated.

Example (6) :

The algebraic measure of the displacement vector of a particle moving in a straight line is given as a function of the time (t in the form

$$s = -t^2 + t$$

Prove that the motion is retarded when $t < 1/2$, and accelerated when $t > 1/2$

Solution :

We can calculate the algebraic measure of the velocity, and the algebraic measure of the acceleration as follows :

$$\begin{aligned}
 v &= \frac{ds}{dt} = -2t + 1 \\
 a &= \frac{dv}{dt} = -2
 \end{aligned}$$

The motion will be retarded when the velocity and acceleration vectors are in opposite directions, i.e. when $v a < 0$

$$\therefore -2(-2t + 1) < 0$$

$$\therefore 2(+2t - 1) < 0$$

$$\therefore t < 1/2$$

In order that the motion will be accelerated, the velocity and acceleration vectors must be in the same direction.

$$\text{i.e. when } va > 0 \qquad \therefore -2(-2t + 1) > 0$$

$$\therefore 2(+2t - 1) > 0 \qquad \therefore t > 1/2$$

Exercise (4)

In the following exercises, $\hat{c}$ is a constant unit vector. $\vec{r}$ is the position vector, $\vec{s}$ is the displacement vector between the initial instant $t = 0$, and the instant t .

- 1) If $\vec{r} = (5t^2 + 1)\hat{c}$, find the derivative $\frac{d\vec{r}}{dt}$ and find its algebraic measure and its magnitude.
- 2) If $\vec{r} = (-t^2 + 2t + 1)\hat{c}$, find $\frac{d\vec{r}}{dt}$ at any time instant, hence find its algebraic measure and its magnitude at $t = 3$.
- 3) If $\vec{r} = (-\frac{3}{2}t^2 + 2t)\hat{c}$, find the displacement vector $\vec{s}$ and find when this vector vanishes. Prove that the motion is retarded when $t < \frac{2}{3}$ and accelerated when $t > \frac{2}{3}$.
- 4) If $\vec{r} = (t^3 + 2)\hat{c}$, find the displacement, velocity and acceleration vectors, hence prove that the motion is accelerated. When will the magnitude of the acceleration be equal to 12 units?
- 5) Find the velocity and acceleration vectors in the following cases, showing which is a uniform, uniformly changing or variable motion :

a) $\vec{r} = t\hat{c}$.	b) $\vec{r} = (2t + 3)\hat{c}$.
c) $\vec{r} = (t^2 + t + 1)\hat{c}$.	d) $\vec{r} = t^3\hat{c}$.
- 6) If $\vec{s} = (2t^3 - t)\hat{c}$, find the velocity and acceleration vectors, and prove that the magnitude of the acceleration is directly

proportional with the time. Show also that the motion will be retarded when $t < \frac{1}{\sqrt{6}}$ and accelerated after that.

7) If $\vec{r} = (t^3 - t^2 - t - 1) \hat{c}$, find the displacement vector, prove that the velocity vanishes when $t = 1$, and that the motion was accelerated when $t = 0$.

8) The algebraic measure of the displacement vector of a particle moving in a straight line as a function of the time is $s = 4.9 t^2 + 2t$. Find the distance described until the instant $t = 3$ and the magnitude of the velocity at this instant. Prove that the motion is accelerated all the time.

9) The algebraic measure of the displacement vector of a particle moving in a straight line as a function of the time is given by the relation $s = -4.9 t^2 + 28 t$. Find the algebraic measures of the velocity and acceleration vectors, and determine when the motion will be retarded and when it will be accelerated ?

10) The algebraic measure of the position vector of a particle moving in a straight line is determined from the relation

$$r = -4.9 t^2 + 19.6 t + 156.8$$

Find the algebraic measures of the displacement, velocity and acceleration vectors. When does the velocity vanish? Find when the motion will be accelerated and when it will be retarded?

The position vector $\vec{r}$ of a body is given by the relation $\vec{r} = (t^3 + bt^2 + 12t + 9) \hat{c}$, where t is the time and $\hat{c}$ is a constant vector. Find the displacement vector and the velocity vector at the instant t . When does the velocity vector vanish ?

General Exercises

[1] Complete :-

- a) $\bar{S} = \bar{r} - \dots$; $\bar{v} = \frac{d.}{dt}$; $\bar{a} = \dots$
- b) If $\bar{r} = (3t - 4)\hat{k}$ then: $\bar{S} = \dots$; $\bar{v} = \dots$; the motion is
- c) If $\bar{S} = (2t^2 + 3t)\hat{k}$ then: $\bar{v} = \dots$; $\bar{a} = \dots$; the motion is
- d) If $\bar{S} = \left(-\frac{1}{2}t^2 + 2t\right)\hat{k}$ then: $\bar{v} = \dots$; $\bar{a} = \dots$; the motion is.....
decelerating when ; and accelerating when
- e) (1) 18 km/h = cm/sec
(2) 54 km/h = m/sec
(3) 20 m/min = m/sec = Cm/sec
(4) 18 km/h/sec = m/sec²
(5) 120 m/min/sec = m/sec² = cm/sec²
(6) 900 m/h/sec = m/sec²
(7) 135 km/min² = m/sec²
- f) If a body falls vertically then : $v = \dots$; $S = \dots$ ($g = 9.8 \text{ m/sec}^2$).

[2] Choose the correct answer from those between brackets :-

- a) If $\bar{r}$ is the position vector of a particle, $\bar{r}_0$ is its initial position vector then
 $\bar{S} = \dots$
 $\left(\frac{d\bar{r}}{dt} \text{ or } \bar{r} - \bar{r}_0 \text{ or } \bar{r}_0 - \bar{r} \text{ or } \bar{r} - \frac{d\bar{r}_0}{dt} \right)$
- b) If $\bar{S} = (2t + 3)\hat{k}$ then the motion is
(uniform or uniformly changing or variable)
- c) If $\bar{S} = (t^3 + 2)\hat{k}$ then the motion is
(uniform or uniformly changing or variable)
- d) If $\bar{S} = (2t^2 + 1)\hat{k}$ then the motion is
(uniform or uniformly changing or variable)
- e) A body moves in a straight line an interval of time t with uniform velocity $\bar{v}$. If its displacement is $\bar{S}$, then $\bar{v} = \dots$
 $\left(\frac{1}{t}\bar{v} \text{ or } \frac{1}{t}\bar{S} \text{ or } \frac{1}{t}\bar{a} \text{ or } \bar{S} \times t \right)$
- f) If a car covers 240 km on a straight road in 180 min. then its average velocity = ...km/h
 $\left(\frac{4}{3} \text{ or } \frac{3}{4} \text{ or } 80 \text{ or } 120 \right)$

g) If a car moves with uniform speed 60 km/h for 20 min. then the distance Covered equals km

(3 or 12 or 20 or 30)

h) If $\vec{v}_A = 12 \hat{k}$ and $\vec{v}_B = 10 \hat{k}$ then $\vec{v}_{AB} = \dots\dots\dots$

($-2 \hat{k}$ or $-22 \hat{k}$ or $2 \hat{k}$ or $22 \hat{k}$)

i) If $\vec{v}_A = 120 \hat{k}$ and $\vec{v}_B = 80 \hat{k}$ then $\vec{v}_{AB} = \dots\dots\dots$

($-40 \hat{k}$ or $-200 \hat{k}$ or $40 \hat{k}$ or $200 \hat{k}$)

[3] Answer the following questions :-

(1) Car A is moving in a straight line, measures the relative velocity of another car B, moving in opposite direction, and found it 120 km/h. When car A had reduced its speed to be $\frac{3}{4}$ of its original, it found that the relative velocity of car B became 100 km/h . Find the actual speed of both cars.

($v_A = 80 \text{ km/h}$; $v_B = 40 \text{ km/h}$)

(2) A car moves on a straight road with a speed 80 km/h. If a motorcycle moves with a speed of 30 km/h. Find the relative velocity of the motorcycle relative to the car if:

a) The motorcycle is moving in the same direction as the car.

b) The motorcycle is moving in opposite direction of the car.

(50 km/h towards the car ; 110 km/h towards the car)

(3) When car A measures the relative velocity of car B, which is moving ahead of car A, it found it 20 km/h. When car A reduced its velocity to the half, it found that the relative velocity of B became 50 km/h. Find the original velocities of the two cars.

($v_A = 60 \text{ km/h}$; $v_B = 80 \text{ km/h}$)

(4) A car moves from Alexandria to Cairo on a straight road. When it is 60 km apart from Cairo, a Truck passed by the car in the direction from Cairo towards Alexandria, and it seemed to the truck driver that the car is moving by 160 km/h. If the speed of the truck is 70 km/h, find in minutes how much time is elapsed from the moment they meet until the car reaches Cairo.

(40 min.)

(5) A cruiser and a ship is moving in the same straight line, towards each other. The cruiser watched the ship, and when it is 20 km apart from it, and moving with 40 km/h the cruiser fired a torpid with a speed 108 km/h. If the speed of the cruiser was 52 km/h at that moment, then find the time elapsed until the torpid targeted the ship.

(6 min)

- (6) The position vector of a particle as a function in time is given by:

$$\vec{r} = (t^2 - 6t + 3)\hat{k}$$

where $\hat{k}$ is a constant unit vector. Find the algebraic measure of the displacement vector and show when the motion is decelerating and when it is accelerating.

(decelerating in $]0,3[$, accelerating in $]3,\infty[$)

- (7) The position vector of a particle as a function in time is given by:

$$\vec{r} = (t + 2)^2 \hat{k}$$

where $\hat{k}$ is a constant unit vector. Find the velocity and acceleration vectors at any time.

($2(t+2) \hat{k}$; $2 \hat{k}$)

- (8) The position vector of a particle as a function in time is given by:

$$\vec{r} = \left(\frac{1}{3}t^3 + 4t - 5 \right) \hat{k}$$

Where $\hat{k}$ is a constant unit vector. Find:

(First) The displacement vector

(2nd) The initial velocity vector

(3rd) The acceleration vector after 5 sec.

$$(\bar{S} = \left(\frac{1}{3}t^3 + 4t \right) \hat{k}; 4\hat{k}; 10\hat{k})$$

- (9) If $\bar{S} = \left(t^3 - \frac{1}{3}t \right) \hat{k}$, then find the vectors of velocity and acceleration and

prove that the acceleration is directly proportion with time, and find when the motion is decelerating and when it is accelerating?

(Decelerating in $]0, \frac{1}{3}[$; accelerating in $] \frac{1}{3}, \infty[$)

- (10) The displacement vector as a function in time is given by

$$\bar{S} = 5\hat{i} + (6t^2 - t^3)\hat{j} \text{ where } \hat{i}, \hat{j} \text{ are two orthogonal unit vectors in the}$$

direction of $\overrightarrow{OX}$ and $\overrightarrow{OY}$. Prove that the motion is accelerating when $t = 1$.

- (11) Find the velocity and acceleration vectors at any moment in each of the following cases, showing when the motion is uniform, variable or uniformly changing:

a) $\vec{r} = (3t - 4)\hat{k}$

b) $\vec{r} = (t^2 + 3t - 5)\hat{k}$

c) $\vec{r} = (2t^3 + 1)\hat{k}$

d) $\vec{r} = \left(\frac{1}{2}t^2 + 3 \right) \hat{k}$

(uniform ; varies uniformly , variable ; varies uniformly)

- (12) If the position vector of a particle is given by $\vec{r} = \left(\frac{2t^3}{27} + \frac{2}{t+2} + 5 \right) \hat{k}$ where

$\hat{k}$ is a constant unit vector, and the magnitude of $\vec{r}$ is in cm, and t in sec. Determine the vectors of displacement, velocity and acceleration, then show when the particle is at instantaneous rest.

- (13) Two cars move on the same straight road in opposite directions. If their velocities are 50 km/h and 70 km/h and the distance between them at a certain moment is 12 km, then find in minutes the time elapsed until the two cars meet.
- (6 min)

- (14) A body begins its motion on a straight line, from rest with uniform acceleration of magnitude 4 cm/sec^2 for 30 sec. It moved with the velocity gained for another 40 sec. Find the magnitude of its average velocity during the whole trip.

($94\frac{2}{7} \text{ cm/sec}$)

- (15) A body moves on a straight line with uniform acceleration. It covers 28.5 cm in the first three seconds, and 64 cm in the ninth and the tenth seconds. Find the acceleration and the initial velocity of the body.

(3 cm/sec^2 ; 5 cm/sec)

- (16) A body moves on a straight line. It covers 52 cm in 4 sec with uniform acceleration, then the acceleration is ceased for 3 sec during which it covers 48 cm after which it moves by a deceleration equals twice its acceleration until it stopped. Find the initial velocity and the total distance covered by the body.

(10 cm/sec ; $142\frac{2}{3} \text{ cm}$)

- (17) A body moves on a straight line with uniform velocity 25 cm/sec. After passing by a certain point by two seconds, another body moved from this point and in the same direction with initial velocity 15 cm/sec and uniform acceleration 8 cm/sec^2 . Find when the two bodies will meet?

(5 sec)

- (18) A body starts its motion in a constant direction with a velocity 15 cm/sec and uniform acceleration 4 cm/sec^2 in the same direction as its velocity.

Find :

(1st) The distance covered in the sixth second only.

(2nd) The distance covered in the seventh and the eighth seconds only.

(37 cm ; 86 cm)

- (19) A body is projected vertically up with a speed 14 m/sec, from a point at a height 350 m above ground. Find the time taken by the body to reach the ground, and the total distance covered by the body. (10 sec ; 370 cm)

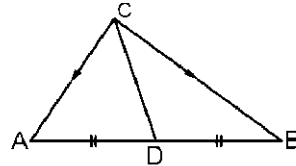
- (20) A rubber ball falls from a height 10 m. It collide with the ground and rebounded vertically up for 2.5 meters. Find the speed of the ball just before and after collision.
(14 m/sec ; 7 m/sec)
- (21) An elevator is at rest at the bottom of a mine. The elevator begins moving up by an acceleration of magnitude 120 cm/sec^2 for 540 cm. then with a uniform velocity for 360 cm, then by a deceleration for 720 cm until it came to rest at the opening of the mine. Find the time taken by the elevator to rise from the bottom to the opening of the mine. (8 sec)
- (22) A stone falls from a height of 90 meters on a pile of sand. It penetrates in it a distance of 14 meters. Find the acceleration of motion inside the sand.
(- 63 m/sec^2)
- (23) From the top of a tower, a body is projected up with a velocity 49 m/sec. The body reached the ground (the tower base) after 12 sec. Find:
(1st) The height of the tower.
(2nd) The maximum height reached by the body above ground.
(117.6 m ; 240.1 m)
- (24) A body is at rest above the ground by 6.125 m, is pulled by a wire up with acceleration 2.45 m/sec^2 . After two seconds from the beginning the wire is cut. Find:
(1st) the speed of the body just before cutting he wire.
(2nd) The maximum height reached by the body above the ground.
(3rd) The velocity of the body when it reaches the ground.
(490 cm/sec ; 12.25 m ; $4.9\sqrt{10} \text{ m/sec}$)

MODEL EXAMS IN MECHANICS

Model [1]**Answer the following questions:****Question [1]: Complete :**

- a) If a body is in equilibrium under the action of three coplanar concurrent forces, then the resultant of any two of these forces is equal in magnitude to and its direction is

- b) In the figure opposite:-
If D is the mid-point of AB then:
 $\overline{CA} + \overline{CB} = \dots\dots\dots$



- c) $72 \text{ km/h} = \dots\dots\dots \text{ m/sec}$
d) Cyclist A moves on a straight road with a speed 10 km/h . If he met another rider B moves with a speed 8 km/h , then the velocity of B with respect to A equals km/h .

Question [2]:

- a) Two forces of magnitude 30 , 16 N , acts on a particle. If the magnitude of their resultant is 26 N , then find the measure of the angle between the forces.
[120°]
b) A particle starts motion with a certain initial speed and an acceleration 0.5 m/sec^2 in the same direction of motion. After covering a distance of 7 m , its speed became 14.4 km/h . Find :
(1st) The initial speed.
(2nd) The time it took to cover the given distance. [3 m/sec ; 2 sec]

Question [3]:

- a) ABCD is a square whose side is 12 cm long. $E \in \overline{BC}$ where $BE = 5 \text{ cm}$. The forces of magnitude 2 , 13 , $4\sqrt{2}$, 9 gm wt act in the directions of $\overline{AB}$, $\overline{AE}$, $\overline{CA}$, $\overline{AD}$ respectively. Find the resultant of these forces.
[$10\sqrt{2} \text{ gm wt}$; 45°]
b) The position vector $\vec{r}$ of a moving particle is given by $\vec{r} = (2t^3 - t + 3)\hat{k}$ where $\hat{k}$ is a constant unit vector, and t is the time. Find the velocity and acceleration vectors then show when the motion is accelerating and where it is decelerating.?

[19.6 m]**Question [4]:**

- a) A weight of magnitude 340 gm wt is suspended by means of two light inextensible strings their lengths are 16 cm and 30 cm attached to two points in the same horizontal level the distance between them is 34 cm . Find the magnitudes of the tensions in each of the two strings. [300 , 160 gm wt]
b) From a window of a house, a small ball is projected vertically up. The ball is shown falling in front of the window after 3 sec from the moment of projection, then it reached the ground after 4 sec from the moment of projection. Find the height of the window above the ground. [19.6 m]

Model [2]

Answer the following questions:

Question [1]: Complete :

- If $\vec{a} = (1, 3)$, $\vec{b} = (2, 1)$ then $\| \vec{a} - \vec{b} \| = \dots\dots\dots$ length units.
- Two forces of magnitude 8 and F gm wt, the measure of the angle between them $\in]0, \pi[$, their resultant bisects the angle between them then F = $\dots\dots\dots$ gm wt.
- If a car moves with uniform speed of 72 km/h for 20 minutes then the distance covered = $\dots\dots\dots$ km.
- If $\vec{v}_a = 12 \hat{k}$; $\vec{v}_b = 8 \hat{k}$ then, $\vec{v}_{ba} = \dots\dots\dots$

Question [2]:

- A body is placed on a smooth incline to the horizontal by an angle of measure 30° . If the body is kept in equilibrium by means of a force $\vec{F}$ whose magnitude is 200 gm wt in direction of the line of greatest slope up. Find the magnitudes of the weight and the reaction of the incline.
[400 , $200\sqrt{3}$ gm wt]
- A car had moved a distance 300 km on a straight road with a speed 100 km/h, then it returned on the same way for 195 km with a speed 65 km/h. Find the average velocity vector of the car for the whole trip. [$17.5 \hat{k}$]

Question [3]:

- In any quadrilateral ABCD prove that : $\vec{AC} + \vec{BD} = \vec{AD} + \vec{BC}$.
- A car applied its brakes when its speed was 45 km/h, to decrease its speed to 18 km/h after it covers 525 m. Find the time taken to cover this distance, and the time taken after that until the car stopped. [60 sec ; 40 sec]

Question [4]:

- A smooth sphere whose radius is 6 cm and its weight is 24 N acts in its center, rests against a vertical smooth wall, and is suspended by means of a string whose length is 4 cm one of its ends is attached to a point on its surface and the other end is fixed at a point of the wall directly above the point of tangency of the sphere with the wall. Find the magnitudes of the tension in the string and the reaction of the wall.
[18 N, 30 N]
- From the top of a hill 9.8 m high, a stone is projected vertically up with a speed 4.9 m/sec. Find :
(1st) The velocity of the stone when it reaches the bottom of the hill. [14.7 m/sec]
(2nd) The time taken to reach the bottom of the hill. [2 sec]

[End of model]

Model [3]

Answer the following questions:

Question [1]: **Complete** :

a) In the figure opposite:-

A force of magnitude $4\sqrt{3}$ N in the direction 30° west of North.

1) The component of the force in the west direction

2) The component of the force in the North direction

b) In the figure opposite:-

A body weighs 90 gm wt is attached to the end of a string of 30 cm long.

The body is pulled by a horizontal Force. It comes to equilibrium when

It is 24 cm apart from the wall AB then:

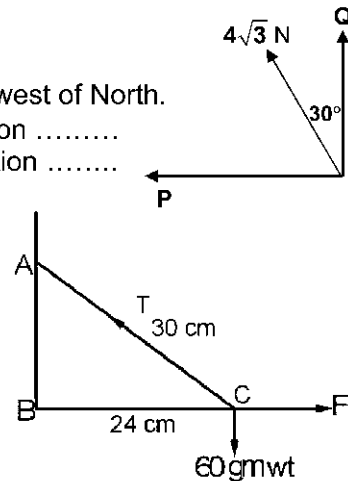
$$1) \frac{F}{\dots} = \frac{T}{\dots} = \frac{90}{\dots}$$

$$2) F = \dots \text{ Gm wt} ; T = \dots \text{ gm wt}$$

c) The displacement vector of a particle is $\vec{S} = (5t^2 + 2t)\hat{k}$, at any moment t . and $\hat{k}$

is a unit vector in the direction of $\vec{S}$, then the magnitude of the velocity of the particle at $t = 2$ equals

d) If a stone falls from the top of a vertical tower, reached the ground after 5 sec, then the height of the tower is m.



Question [2]:

a) Two forces of magnitudes 12, 15 N act on a particle, the angle between them is

θ° where $\tan \theta = \frac{-3}{4}$. Find the magnitude of their resultant and the measure of

the angle between the resultant and the first force. [9 N , 90°]

b) A ship moves on a straight line towards a port. When it is at 100 km from it, an airplane bypassed above the ship in opposite direction with speed 250 km/h. If the plane observed that the ship is moving with 300 km/h, then find the time elapsed from the moment of observation until the ship lands at the port [2 hours]

Question [3]:

a) A uniform rod whose length is 80 cm and its weight is 12 N acts in its mid-point. The rod is freely suspended from its ends by means of two strings then the other ends are attached to a fixed nail in the ceiling. If the two strings are perpendicular and one of them is 48 cm. Find in equilibrium the magnitude of the tensions in the two strings. [7.2N, 9.6 N]

b) A body moved from rest with uniform acceleration of 3 cm/sec^2 until it reached a velocity 60 cm/sec. The acceleration then ceased and the body continued with the velocity gained an interval of time, then it moved with uniform deceleration its magnitude is 12 cm/sec^2 until it came to rest. If the total distance covered is 1050 cm, then find the time of acceleration and that of deceleration then find the average speed for the whole motion, [$t_1 = 20 \text{ sec}$; $t_2 = 5 \text{ sec}$; 350 cm/sec]

Question [4]:

a) Four coplanar and concurrent forces act on a particle. The first is 4 N, and acts due to East, the second is 2 N and acts at 30° North East, the third is 5 N and acts in direction 60° North West and the fourth is $3\sqrt{3}$ N in direction 60° West south. Find the magnitude and direction of the resultant of these forces.

[4 N ; 120°]

b) A particle is projected vertically up from a point on the ground. If the maximum height can be attained by the particle is 44.1 m, then find:

(1st) The velocity of projection. [29.4 m/sec]

(2nd) The time taken, from the moment of projection, until it return to the ground surface. [6 sec]

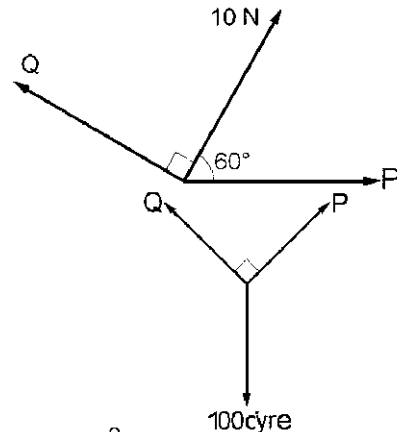
[End of model]

Model [4]

Answer the following questions:

Question [1]: Complete :

- a) In the figure opposite :-
If the force 10 N is resolved into two components P and Q inclined to the force by 60° and 90° then: $P = \dots\dots$ N, $Q = \dots\dots$ N



- b) In the figure opposite:-
A light inextensible string passes through a smooth ring of weight 100 Dynes, then
 $T_1 = \dots\dots$ dyne ; $T_2 = \dots\dots$ dyne.
- c) A body moves on a straight line decelerates by 4 m/sec^2 . If it stopped after 15 sec, then the magnitude of its initial velocity equals $\dots\dots\dots$ m/sec.
- d) If $\vec{r} = (t^3 + t^2 + 5t - 3)\hat{k}$ is the position vector of a particle moving on a straight line, then the velocity vector of this particle is $\vec{v} = \dots\dots\dots$

Question [2]:

- a) Two forces of magnitude F and 4 N, act on a particle. The angle between them measures 120° . If the magnitude of their resultant is $4\sqrt{3}$ N, then find:
(1st) the value of F and the measure of the angle between the resultant and the first force. [8 N , 30°]
(2nd) The magnitude of the minimum and maximum values of the resultant. [$4 \leq R \leq 12$]
- b) The position vector of a moving particle at any time t sec, is given by $\vec{r} = (5 + 16t - 2t^2)\hat{k}$, where S in cm. Find the vectors of displacement, velocity and acceleration, then find:
(1st) The displacement vector at the moment when the velocity vanishes. [$32 \hat{k}$]
(2nd) The velocity vector when the displacement vanishes. [$\pm 16 \hat{k}$]

Question [3]:

- a) If $\vec{b} = (-2, 6)$, $\vec{c} = (4, -1)$, $2\vec{a} - 3\vec{b} = 4\vec{c}$. Find $\|\vec{a}\|$. [$\sqrt{74}$]
- b) A rubber ball fall from a height of 19.6 m on the surface of a horizontal solid ground. If the ball rebound up, then find the maximum height the ball will reach, given that the ball had lost half its speed due to impact. [4.9 m]

Question [4]:

- a) The coplanar concurrent forces of magnitudes 5,4,P,3,Q and 7 kg wt are applied on a particle. The angle between each two successive forces is 60° . Find the magnitudes of P and Q such that the system is in equilibrium.
[$P = 9 \text{ kg wt}$, $Q = 6 \text{ kg wt}$]

Mechanics

- b)** A body moves in constant direction with constant acceleration. If the body covers 20 m in the third second from the beginning of motion and 150 m during the eighth, ninth and tenth seconds. Find the magnitude of the acceleration and its initial velocity. [5 m/sec² , 7.5 m/sec]
-

[End of model]

Answers of Statics

Exercise (1)

- 4) (4 , -8)
- 5) $\sqrt{13}$, $\sqrt{13}$, 3 and 4
- 6) $\sqrt{10}$

Exercise (2)

- 1) $15\sqrt{3}$ newtons, 15 newtons
- 2) $20\sqrt{3}$ newtons, 20 newtons
- 3) $F_1 = \frac{500 \sin 45^\circ}{\sin 75^\circ} \simeq 366$ newtons
 $F_2 = \frac{500 \sin 60^\circ}{\sin 75^\circ} \simeq 448.3$ newtons
- 4) 2 , 1 , 2.7 , F newtons
- 5) 10 , $10\sqrt{3}$
- 6) $50\sqrt{2}$, $50\sqrt{2}$

Exercise (3)

- 1) 4 newtons
- 2) 2 F where F is the magnitude of each force
- 3) 5 newtons
- 4) 3.93 newtons
- 5) $R \simeq 15.58$ newtons $\theta = 43.21$ with $\vec{AO}$
- 6) $R = \sqrt{13}$ newtons, $\theta = 56.19$

Exercise (4)

- 1) 6 newtons, 2.5 newtons
- 2) $138^{\circ} 36'$
- 3) $19^{\circ} 28'$, 18.24 newtons
- 4) The tension = 30 newtons, the reaction = $30\sqrt{3}$ newtons
- 5) The force = $\frac{1}{2}$ newton, the tension = $\frac{\sqrt{3}}{2}$ newtons
- 6) The string form $\tan^{-1} \frac{3}{4}$ with the vertical, tension in AB, BC=500, 300 gm. wt. respectively.
- 7) 21 gm wt, 75 gm wt., 72 gm wt.

Exercise (5)

- 1) The pressure on the wall = $5\sqrt{3}$ newtons, the tension in the string = $10\sqrt{3}$ newtons.
- 2) The pressure on each rod = $\frac{10\sqrt{3}}{3}$ newtons
- 3) The pressure on the wall = $30\sqrt{3}$ newtons, the pressure on the plane = 60 newtons.
- 4) The inclination of the rod with the vertical = 43.43° . The reaction at the first end = $\frac{\sqrt{5}}{2}$ times the weight of the rod

- 5) The tension in the two strings 10, 24 newtons
- 6) The rod form with the horizontal 30°
- 7) 2, $2\sqrt{3}$ newtons, 30°
- 8) 30° , $\frac{\sqrt{3}}{2}$ times the weight.

Answers of Dynamics

Exercise (1)

- 1) $\frac{25}{2} \hat{k}$
- 2) 82.6 km/h.
- 3) 80 km/h.
- 4) 2 hours
- 5) 40 km/h, 100 km/h.
- 6) 250 metre / second
- 7) 2 minutes

Exercise (2)

- 1) 2m/sec^2 , 3m/sec
- 2) 70 sec, 1078 cm.
- 3) $55\frac{2}{3}$ sec.
- 4) 13212 m, 19.515 m/sec.
- 5) 880 m.
- 6) 1 minute, 1400 m.
- 7) 60 m/sec, 360 m.
- 8) $5 \times 10^7 \text{ cm/sec}^2$, 100 cm/sec.
- 9) 1 m./sec^2 , $-\frac{25}{12} \text{ m/sec}^2$
- 10) 2 cm/sec^2 , 1775 cm.

Exercise (3)

- 1) 39.2 m/sec, $\sqrt{14}$ sec
- 2) 27.6 m, 23.6 m/sec.
- 3) 2 sec., 19.6 m.
- 4) 5 sec.
- 5) 1 sec during ascending, 3 sec during descending, 9.8 m/sec in ascending -9.8 m/sec in descending.
- 6) 10 sec.
- 7) 8 sec., 58.8 m/sec.
- 8) 24.5 m.

Exercise (4)

- 1) $(10t) \hat{c}$, the algebraic measure = magnitude = $10t$.
- 2) $(-2t + 2) \hat{c}$, -4, 4
- 3) $\vec{s} = \vec{r}$ and it vanishes when $t = 0, \frac{4}{3}$
- 4) $\vec{s} = t^3 \hat{c}$, $\vec{v} = 3t^2 \hat{c}$, $\vec{a} = 6t \hat{c}$, $t = 2$.
- 5) a) $\vec{v} = \vec{c}$, $\vec{a} = \vec{0}$ uniform motion.
 b) $\vec{v} = 2\hat{c}$, $\vec{a} = \vec{0}$ uniform motion
 c) $\vec{v} = (2t + 1) \hat{c}$, $\vec{a} = 2 \hat{c}$, motion uniformly changing.
 d) $\vec{v} = 3t^2 \hat{c}$, $\vec{a} = 6t \hat{c}$, changing motion.

Mechanics

6) $\vec{v} = (6t^2 - 1) \hat{c}$, $\vec{a} = 12t \hat{c}$

7) $\vec{s} = (t^3 - t^2 - t) \hat{c}$

8) 50.1, 31.4

9) $v = -9.8t + 28$, $a = -9.8$, the motion is a retardation when $t < \frac{20}{7}$,
and accelerated when $t > \frac{20}{7}$

10) $s = -4.9t^2 + 19.6t$, $v = -9.8t + 19.6$

$a = -9.8$, $t = 2$, the motion is a retardation when $t < 2$, and
accelerated when $t > 2$.

11) $t = 2$

